TayChenJiangNguyenLi-stat443project

TayChenJiangNguyenLi

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Loading packages

```
library(readr)
library(dplyr)
library(lubridate)
```

Reading the data

```
# Cover 1981-01 to 2025-12
nhpi_full <- read.csv("18100205.csv", na.strings = c("..", "NA", "")) %>%
  mutate(REF_DATE = as.Date(paste0(REF_DATE, "-01")),
         REF_DATE_ym = format(REF_DATE, "%Y-%m"))
# head(nhpi_full)

# Cover 1981-01 to 2023-12
ir <- read_csv("IRSTCB01CAM156N.csv",show_col_types = FALSE) %>%
  rename(REF_DATE = observation_date,
         IR = IRSTCB01CAM156N) %>%
  mutate(REF_DATE = as.Date(REF_DATE),
         REF_DATE_ym = format(REF_DATE, "%Y-%m"),
         IR = as.numeric(IR)) %>%
  filter(!is.na(IR))
# head(ir)

# Combine dataset for ARMAX (cover the overlap only)
nhpi_ir <- nhpi_full %>%
  left_join(ir %>% select(REF_DATE_ym, IR),
            by = "REF_DATE_ym") %>%
  filter(REF_DATE <= as.Date("2023-12-01"))
# head(nhpi_ir)

# Lagged interest rate
nhpi_ir <- nhpi_ir %>%
  group_by(GEO, New.housing.price.indexes) %>%
  arrange(REF_DATE) %>%
  mutate(IR_lag1 = lag(IR, 1),
         IR_lag2 = lag(IR, 2)) %>%
  ungroup()
```

```r
# head(nhpi_ir)

# Checking date coverage
range(nhpi_full$REF_DATE)  # should include 1981-01-01 to 2025-12-01 (or close)
range(ir$REF_DATE)         # should include 1981-01-01 to 2023-12-01
range(nhpi_ir$REF_DATE)    # should include 1981-01-01 to 2023-12-01

# Any missing IR after merge? (should be basically none within overlap)
mean(is.na(nhpi_ir$IR))

# Check lags (first month per series will be NA)
nhpi_ir %>%
  filter(GEO == "Canada", New.housing.price.indexes == "Total (house and land)") %>%
  select(REF_DATE, VALUE, IR, IR_lag1, IR_lag2) # %>%
  # head(5)
```

Defining functions

Baseline 1: Naive/Persistence

```r
persist_fc <- function(train, holdout, iprint = FALSE) {
  n <- length(train)
  n_holdout <- length(holdout)
  mse <- 0
  fcvec <- numeric(n_holdout)

  # first forecast
  fc <- train[n]
  yt <- holdout[1]
  fcvec[1] <- fc
  fcerror <- yt - fc
  mse <- mse + fcerror^2

  # remaining forecasts
  for (i in 2:n_holdout) {
    yt <- holdout[i]
    fc <- holdout[i-1]
    fcvec[i] <- fc
    fcerror <- yt - fc
    mse <- mse + fcerror^2
  }

  if (iprint)
    print(cbind(holdout, fcvec))

  rmse <- sqrt(mse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

Baseline 2: Training-Set Average

```
iid_fc <- function(train, holdout, iprint = FALSE) {
  n <- length(train)
  n_holdout <- length(holdout)
  average <- mean(train)
  mse <- 0
  fcvec <- numeric(n_holdout)

  for (i in 1:n_holdout) {
    yt <- holdout[i]
    fc <- average
    fcvec[i] <- fc
    fcerror <- yt - fc
    mse <- mse + fcerror^2
  }

  if (iprint)
    print(cbind(holdout, fcvec))

  rmse <- sqrt(mse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

Method 1: Best Exponential Smoothing (allowing for trend and seasonality, if relevant)

```
# Simple Exponential Smoothing
esm_fc <- function(train, holdout, alpha, level, iprint = FALSE) {
  n_holdout <- length(holdout)
  sse <- 0
  fcvec <- numeric(n_holdout)

  fc <- level
  yt <- holdout[1]
  level_new <- level
  fcvec[1] <- fc
  fcerror <- yt - fc
  sse <- sse + fcerror^2

  for (i in 2:n_holdout) {
    level_new <- alpha * holdout[i-1] + (1 - alpha) * level_new

    fc <- level_new
    fcvec[i] <- fc
    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + fcerror^2

    if (iprint) {
      cat("i =", i, "| fc =", fc, "| y =", yt, "| level =", level_new, "\n")
    }
```

```r
  }

  rmse <- sqrt(sse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

```r
# Linear Exponential Smoothing
lholt_fc <- function(train, holdout, alpha, beta, level, slope, iprint = FALSE) {
  n_holdout <- length(holdout)
  sse <- 0
  fcvec <- numeric(n_holdout)

  # first forecast
  fc <- level + slope
  yt <- holdout[1]
  newfc <- fc
  fcvec[1] <- fc
  fcerror <- yt - fc
  sse <- sse + fcerror^2

  level_prev <- level
  slope_prev <- slope

  if (iprint) {
    cat("i = 1 | fc =", fc, "| y =", yt, "| level =", level_prev, "| slope =", slope_prev, "\n")
  }

  # remaining forecasts
  for (i in 2:n_holdout) {
    level_new <- alpha * holdout[i-1] + (1 - alpha) * fc
    slope_new <- beta * (level_new - level_prev) + (1 - beta) * slope_prev

    fc <- level_new + slope_new
    fcvec[i] <- fc
    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + fcerror^2

    if (iprint) {
      cat("i =", i, "| fc =", fc, "| y =", yt, "| level =", level_new, "| slope =", slope_new, "\n")
    }

    level_prev <- level_new
    slope_prev <- slope_new
  }

  rmse <- sqrt(sse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

```r
# Additive Seasonal Exponential Smoothing
aseason_fc <- function(train, holdout, alpha, beta, gamma, level, slope, season, iprint = FALSE) {
  n_holdout <- length(holdout)
```

```r
m <- length(season)
sse <- 0
fcvec <- numeric(n_holdout)

# prev states from training end
level_prev <- level
slope_prev <- slope

# helper: map i -> seasonal position in 1..d
spos <- function(i) ((i-1) %% m) + 1

# First forecast
fc <- level_prev + slope_prev + season[spos(1)]
fcvec[1] <- fc
yt <- holdout[1]
fcerror <- yt - fc
sse <- sse + fcerror^2

if (iprint) {
  cat("i = 1 | fc =", fc, "| y =", yt, "| level =", level_prev, "| slope =", slope_prev, "| season ="
}

# Remaining forecast
for (i in 2:n_holdout) {
  # Seasonal of previous observation (i-1)
  k_prev <- spos(i-1)
  s_prev <- season[k_prev]
  # Update using previous observation holdout[i-1]
  level_new <- alpha * (holdout[i-1] - s_prev) + (1 - alpha) * (level_prev + slope_prev)
  slope_new <- beta * (level_new - level_prev) + (1 - beta) * slope_prev
  s_new <- gamma * (holdout[i-1] - level_new) + (1 - gamma) * s_prev

  # store updated seasonal state one full cycle later
  season[k_prev] <- s_new

  # Seasonal of current forecast (i)
  k_cur <- spos(i)
  s_cur <- season[k_cur]

  # Forecast current observation using updated states + current seasonal
  fc <- level_new + slope_new + s_cur
  fcvec[i] <- fc
  yt <- holdout[i]
  fcerror <- yt - fc
  sse <- sse + fcerror^2

  if (iprint) {
    cat("i =", i, "| fc =", fc, "| y =", yt, "| level =", level_new, "| slope =", slope_new, "| seaso
  }

  # Update level and slope
  level_prev <- level_new
  slope_prev <- slope_new
```

```
  }

  rmse <- sqrt(sse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

Method 2: ARIMA

```
arma_fc <- function(tsdata, ntrain, order, seasonal, method, traincoef, include.mean, iprint=F) {
  obj = arima(tsdata, order=order, seasonal = seasonal, init=traincoef, fixed=traincoef,
              method=method, include.mean=include.mean, optim.control=list(maxit=0)) # no log-likelihoo
  fc = tsdata - obj$residuals

  ntotal = length(tsdata)
  holdout_fc = fc[(ntrain+1):ntotal]
  holdout = tsdata[(ntrain+1):ntotal]

  if(iprint) print(cbind(holdout,holdout_fc))

  rmse = sqrt(mean((holdout-holdout_fc)^2))
  return(list(rmse=rmse, fc=holdout_fc, residuals = obj$residuals))
}
```

Method 3: ARIMAX with lagged interest

```
armax_fc <- function(tsdata, ntrain, order, seasonal, xreg, method, traincoef, include.mean, iprint=F) {
  obj = arima(tsdata, order=order, seasonal = seasonal, init=traincoef,
              fixed=traincoef, method=method, include.mean=include.mean,
              xreg=xreg, optim.control=list(maxit=0))

  fc = tsdata - obj$residuals

  ntotal = length(tsdata)
  holdout_fc = fc[(ntrain+1):ntotal]
  holdout = tsdata[(ntrain+1):ntotal]

  if(iprint) print(cbind(holdout,holdout_fc))

  rmse = sqrt(mean((holdout-holdout_fc)^2))
  return(list(rmse=rmse, fc=holdout_fc, residuals = obj$residuals))
}
```

Analyzing Montreal

Setting up

```r
# Reading the data
montreal_nhpi_data <- read.csv("Full_Cities/montreal_full.csv")
montreal_nhpi_ir_data <- read.csv("Extracted_Cities/montreal.csv")

# Extracting the NHPI
montreal_nhpi <- montreal_nhpi_data$VALUE

# Extracting the one-month lag interest rate
montreal_irlag1 <- montreal_nhpi_ir_data$IR_lag1

# Setting the time window from 1981-2023
montreal_ts <- ts(montreal_nhpi, start = c(1981,1), end = c(2023, 12), frequency = 12)
montreal_ir_ts <- ts(montreal_irlag1, start = c(1981,1), end = c(2023, 12), frequency = 12)

# Applying a transformation for NHPI
montreal_transformed_nhpi <- diff(log(montreal_ts))

# Setting a training set from 1981-2021
montreal_train_ts <- window(montreal_transformed_nhpi, end = c(2021,12))
montreal_train_ir_xreg <- window(montreal_ir_ts, start = c(1981,2), end = c(2021,12))

# Setting a holdout set from 2022-2023
montreal_holdout_ts <- window(montreal_transformed_nhpi, start = c(2022,1))
montreal_holdout_ir_xreg <- window(montreal_ir_ts, start = c(2022,1))
```

Doing the baseline forecasts

```r
# Baseline 1: Naive/Persistence
montreal_persist_fc <- persist_fc(montreal_train_ts, montreal_holdout_ts, iprint = F)

# Baseline 2: Training-Set Average
montreal_iid_fc <- iid_fc(montreal_train_ts, montreal_holdout_ts, iprint = F)
```

Doing the exponential smoothing forecasts

```r
# Method 1
ntrain <- length(montreal_train_ts)
ntotal <- length(montreal_ts)-1

montreal_transformed_nhpi_train <- montreal_transformed_nhpi[1:ntrain]
montreal_transformed_nhpi_holdout <- montreal_transformed_nhpi[(ntrain+1):ntotal]

# Simple Exponential Smoothing
fit_expsmo <- HoltWinters(montreal_train_ts, beta = F, gamma = F)
montreal_esm_fc <- esm_fc(montreal_transformed_nhpi_train, montreal_transformed_nhpi_holdout,
                          alpha = fit_expsmo$alpha, level = fit_expsmo$coefficients["a"],
                          iprint = F)

# Linear Exponential Smoothing
fit_hw <- HoltWinters(montreal_train_ts, gamma = F)
```

```r
montreal_holt_fc <- lholt_fc(montreal_transformed_nhpi_train, montreal_transformed_nhpi_holdout,
                             alpha = fit_hw$alpha, beta = fit_hw$beta,
                             level = fit_hw$coefficients["a"],
                             slope = fit_hw$coefficients["b"], iprint = F)

# Additive Seasonal Exponential Smoothing
wafit <- HoltWinters(montreal_train_ts, seasonal = "additive")
montreal_aseason_fc <- aseason_fc(montreal_transformed_nhpi_train,
                                  montreal_transformed_nhpi_holdout,
                                  alpha = wafit$alpha, beta = wafit$beta,
                                  gamma = wafit$gamma, level = wafit$coef[1],
                                  slope = wafit$coef[2], season = wafit$coefficient[3:14],
                                  iprint = F)

# Comparing RMSE
rmse_esm <- cbind(montreal_esm_fc$rmse, montreal_holt_fc$rmse, montreal_aseason_fc$rmse)
colnames(rmse_esm) <- c("esm", "lholt", "aseason") # RMSE: 0.0079, 0.0077, 0.0088
rmse_esm
```

```
##           esm       lholt     aseason
## a 0.007918044 0.007662966 0.008806579
```

```r
# Best exponential smoothing method: Linear Exponential Smoothing
```

Doing the ARIMA forecasts

```r
# Method 2
montreal_transformed_nphi_auto <- forecast::auto.arima(montreal_train_ts)
```

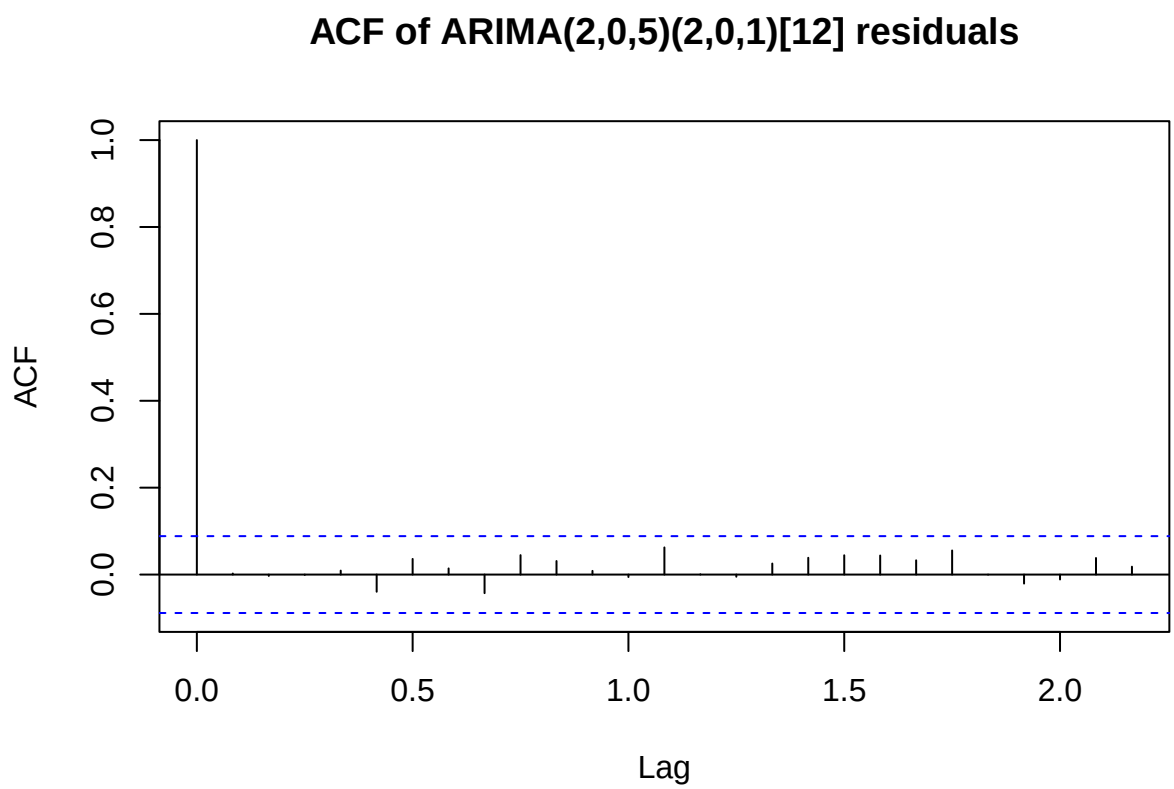
```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
summary(montreal_transformed_nphi_auto) # 0.0048783
```

```
## Series: montreal_train_ts
## ARIMA(2,0,5)(2,0,1)[12] with non-zero mean
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2933  -0.4683  -1.1324  0.3576  0.0775  -0.0112  0.1526  0.7129  0.0270
## s.e.  0.1615   0.1643   0.1629  0.1522  0.0680   0.0678  0.0628  0.2571  0.0723
##          sma1    mean
##       -0.6604  0.0034
## s.e.   0.2622  0.0007
##
## sigma^2 = 2.434e-05:  log likelihood = 1916.27
## AIC=-3808.54   AICc=-3807.89   BIC=-3758.19
##
## Training set error measures:
```

```
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -4.22165e-05 0.004878312 0.002960714 NaN  Inf 0.7689245
##                     ACF1
## Training set 0.002456102
```

```r
resid_fit <- residuals(montreal_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(2,0,5)(2,0,1)[12] residuals")
```




```r
# Comparing with the other ARIMA models
manual_fit <- arima(montreal_train_ts, order = c(2,0,5),
                    seasonal = list(order = c(2,0,1), period = 12), method = "ML")
```

```
## Warning in log(s2): NaNs produced
```

```
## Warning in log(s2): NaNs produced
```

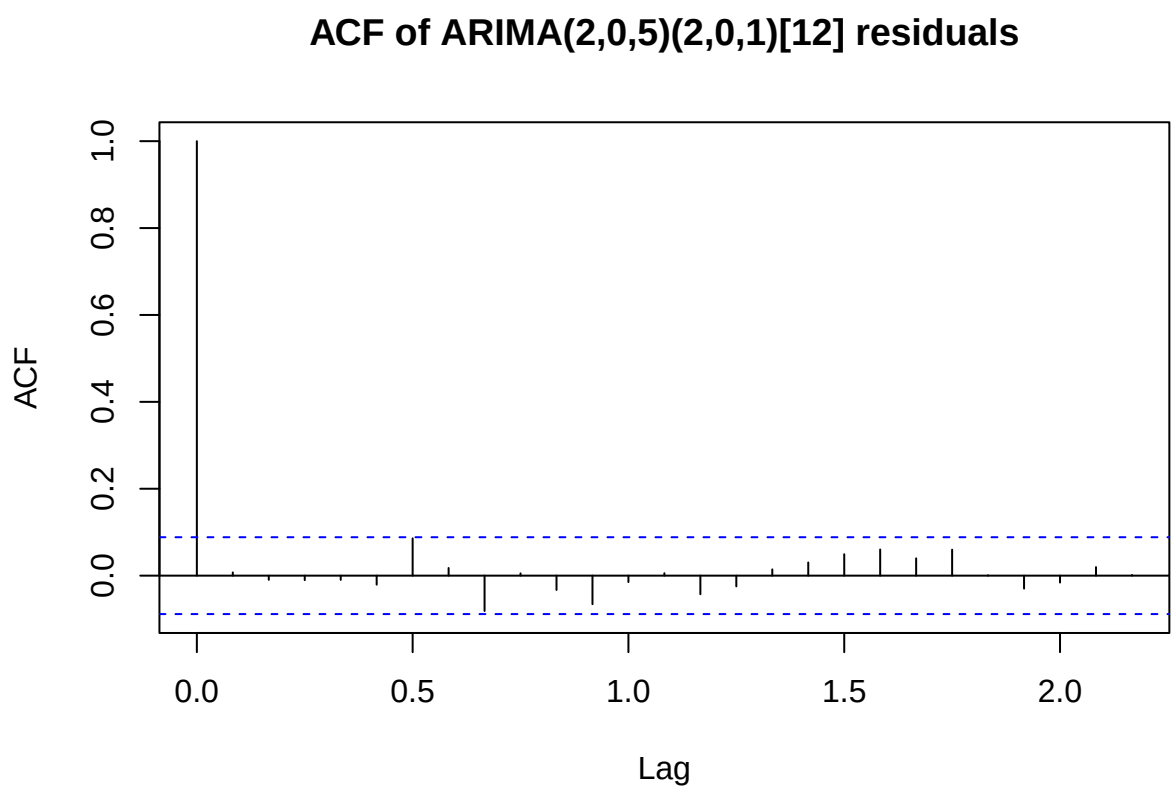
```r
summary(manual_fit) # RMSE: 0.004903117
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 5), seasonal = list(order = c(2,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
```

```
## Warning in sqrt(diag(x$var.coef)): NaNs produced

##          ar1     ar2      ma1      ma2    ma3      ma4     ma5    sar1    sar2
##       0.8764  0.0595  -0.7152  -0.1025  0.026  -0.0082  0.0892  0.7867  0.0841
## s.e.     NaN     NaN   0.0202      NaN    NaN   0.0262     NaN  0.0015  0.0022
##          sma1  intercept
##       -0.8241     0.0042
## s.e.   0.0429     0.0013
##
## sigma^2 estimated as 2.404e-05:  log likelihood = 1913.85,  aic = -3803.7
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE     MASE        ACF1
## Training set -0.000121822 0.004903117 0.002922884 NaN  Inf 0.772463 0.007374584
```

```
acf(residuals(manual_fit), main = "ACF of ARIMA(2,0,5)(2,0,1)[12] residuals")
```




```
test1 <- arima(montreal_train_ts, order = c(2,0,4),
               seasonal = list(order = c(2,0,1), period = 12), method = "ML")
summary(test1) # RMSE: 0.004919859
```

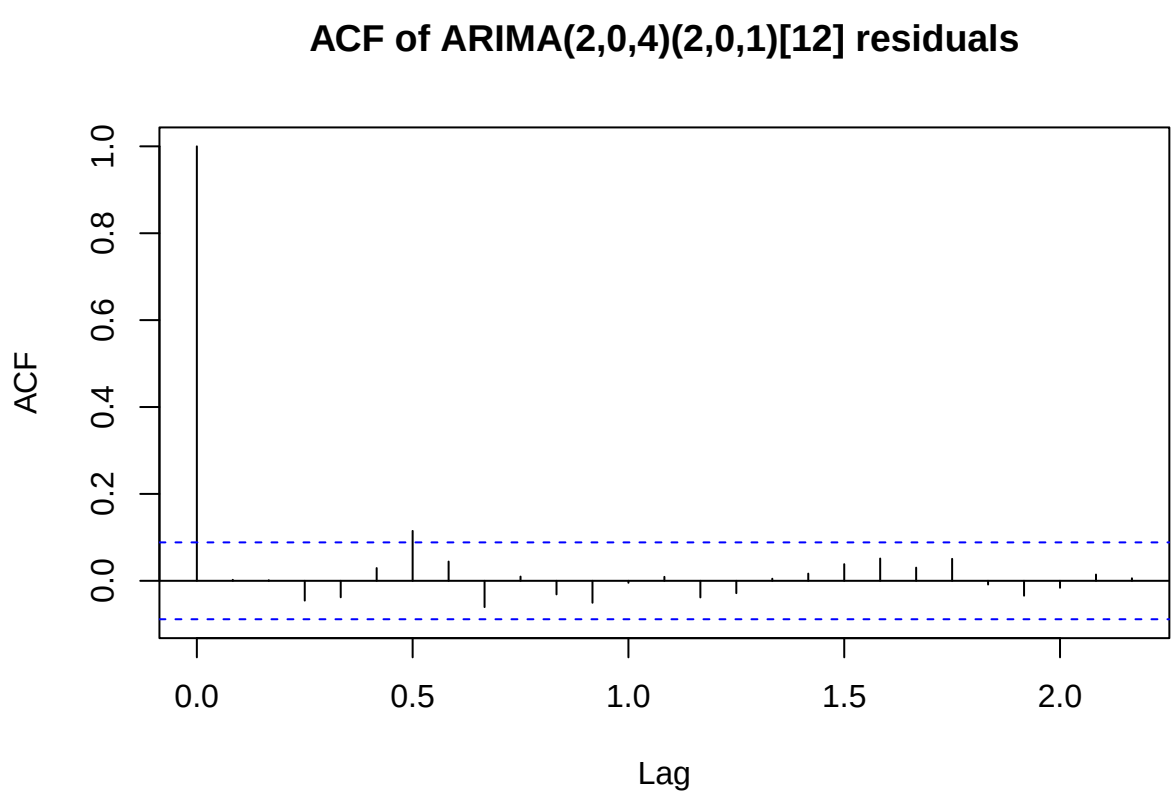
```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 4), seasonal = list(order = c(2,
```

```
##     0, 1), period = 12), method = "ML")
##
## Coefficients:

## Warning in sqrt(diag(x$var.coef)): NaNs produced

##          ar1     ar2     ma1      ma2      ma3     ma4    sar1    sar2     sma1
##       0.1009  0.8396  0.0708  -0.7734  -0.0067  0.0520  0.8286  0.0792  -0.8625
## s.e.     NaN     NaN     NaN      NaN   0.0474  0.0536  0.1540  0.0648   0.1467
##       intercept
##          0.0043
## s.e.     0.0018
##
## sigma^2 estimated as 2.421e-05:  log likelihood = 1911.91,  aic = -3801.82
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -8.718343e-05 0.004919859 0.002886015 NaN  Inf 0.7627192
##                    ACF1
## Training set 0.00222566
```

```r
acf(residuals(test1), main = "ACF of ARIMA(2,0,4)(2,0,1)[12] residuals")
```



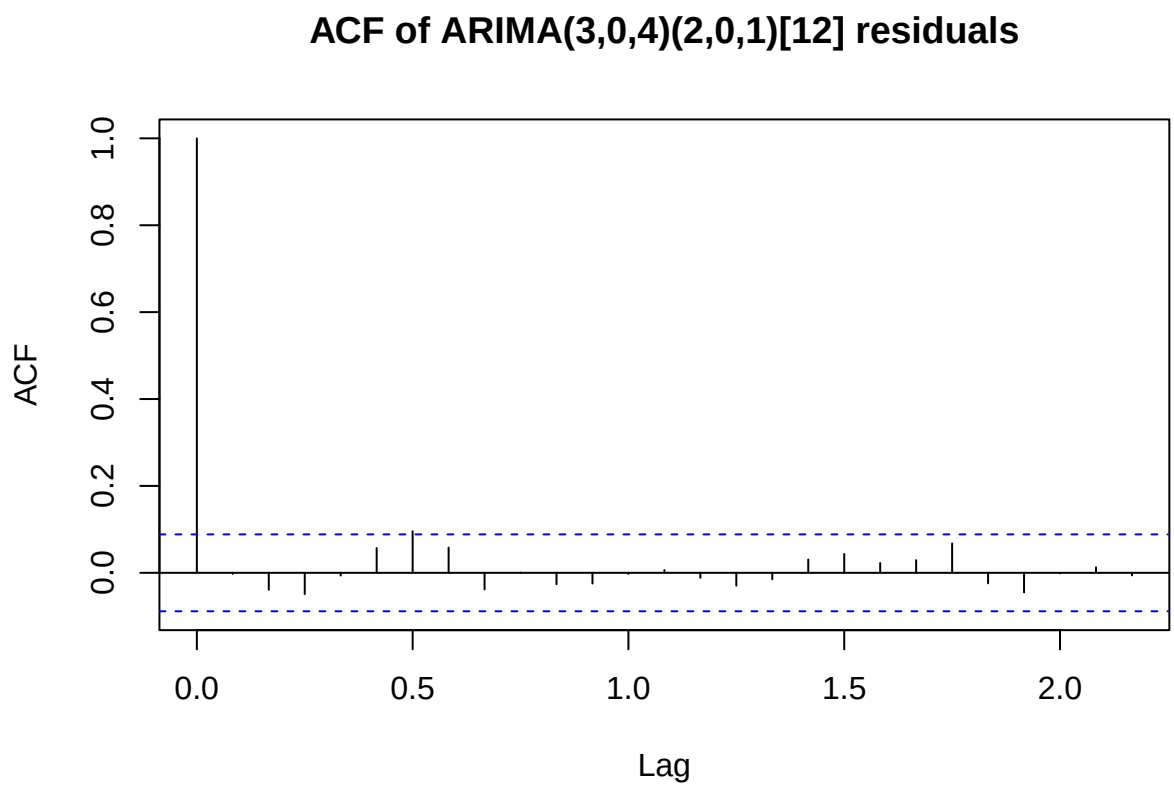
```r
test2 <- arima(montreal_train_ts, order = c(3,0,4),
               seasonal = list(order = c(2,0,1), period = 12), method = "ML")
```

```
## Warning in arima(montreal_train_ts, order = c(3, 0, 4), seasonal = list(order =
## c(2, : possible convergence problem: optim gave code = 1
```

```r
summary(test2) # RMSE: 0.004857711
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(3, 0, 4), seasonal = list(order = c(2,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2     ar3      ma1     ma2      ma3      ma4    sar1
##       0.2633  -0.2965  0.9676  -0.0730  0.3999  -0.8139  -0.0725  0.8008
## s.e.  0.0164   0.0139  0.0167   0.0509  0.0328   0.0344   0.0486  0.1765
##         sar2     sma1  intercept
##       0.1004  -0.8507     0.0046
## s.e.  0.0677   0.1709     0.0021
##
## sigma^2 estimated as 2.36e-05:  log likelihood = 1915.58,  aic = -3807.16
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE     MASE
## Training set -9.30943e-05 0.004857711 0.002842068 NaN  Inf 0.751105
##                      ACF1
## Training set -0.002624896
```

```r
acf(residuals(test2), main = "ACF of ARIMA(3,0,4)(2,0,1)[12] residuals")
```




```r
test3 <- arima(montreal_train_ts, order = c(3,0,5),
               seasonal = list(order = c(2,0,1), period = 12), method = "ML")
```

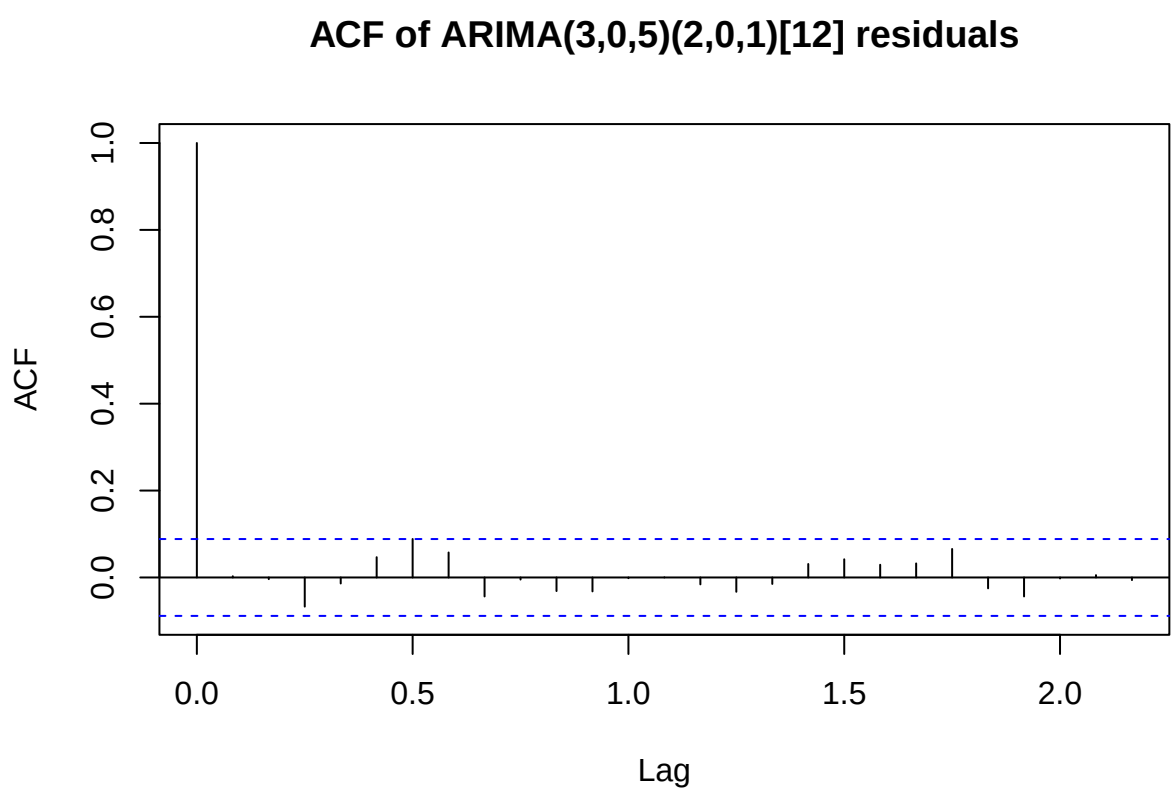
```
## Warning in arima(montreal_train_ts, order = c(3, 0, 5), seasonal = list(order =
## c(2, : possible convergence problem: optim gave code = 1
```

```r
summary(test3) # RMSE: 0.004852676
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(3, 0, 5), seasonal = list(order = c(2,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2     ar3      ma1     ma2      ma3      ma4     ma5
##       0.2642  -0.2936  0.9666  -0.0852  0.3585  -0.7956  -0.0706  0.0531
## s.e.  0.0164   0.0137  0.0184   0.0494  0.0486   0.0421   0.0472  0.0536
##         sar1    sar2     sma1  intercept
##       0.8061  0.1085  -0.8711     0.0046
## s.e.  0.1369  0.0645   0.1273     0.0022
##
## sigma^2 estimated as 2.355e-05:  log likelihood = 1916.07,  aic = -3806.15
##
## Training set error measures:
```

```
##                          ME        RMSE         MAE MPE MAPE      MASE
## Training set -8.857333e-05 0.004852676 0.002829046 NaN  Inf 0.7476633
##                     ACF1
## Training set 0.002781505
```

```r
acf(residuals(test3), main = "ACF of ARIMA(3,0,5)(2,0,1)[12] residuals")
```




```r
test4 <- arima(montreal_train_ts, order = c(2,0,5),
               seasonal = list(order = c(2,0,2), period = 12), method = "ML")
summary(test4) # RMSE: 0.004908301
```

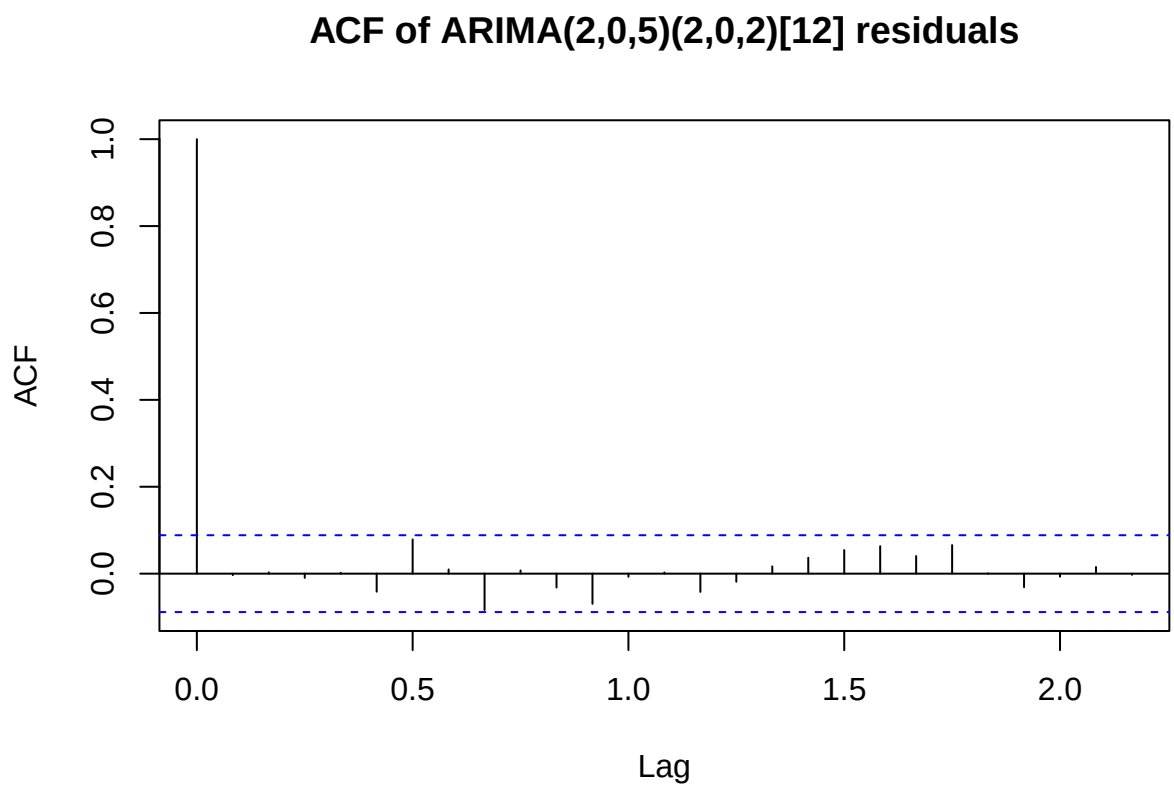
```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 5), seasonal = list(order = c(2,
##     0, 2), period = 12), method = "ML")
##
## Coefficients:

## Warning in sqrt(diag(x$var.coef)): NaNs produced

##          ar1     ar2      ma1      ma2     ma3      ma4     ma5    sar1    sar2
##       0.8518  0.0693  -0.6860  -0.1222  0.0409  -0.0209  0.1239  0.6627  0.2625
## s.e.  0.0011  0.0011   0.0028      NaN     NaN      NaN  0.0018  0.0014  0.0017
##          sma1     sma2  intercept
```

```
##       -0.7106  -0.1882     0.0043
## s.e.   0.0017   0.0025     0.0011
##
## sigma^2 estimated as 2.409e-05:  log likelihood = 1913.66,  aic = -3801.32
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001793343 0.004908301 0.002963063 NaN  Inf 0.7830814
##                      ACF1
## Training set -0.003464896
```

```
acf(residuals(test4), main = "ACF of ARIMA(2,0,5)(2,0,2)[12] residuals")
```




```
test5 <- arima(montreal_train_ts, order = c(2,0,4),
               seasonal = list(order = c(2,0,2), period = 12), method = "ML")
```

```
## Warning in arima(montreal_train_ts, order = c(2, 0, 4), seasonal = list(order =
## c(2, : possible convergence problem: optim gave code = 1
```

```
summary(test5) # RMSE: 0.004891118
```

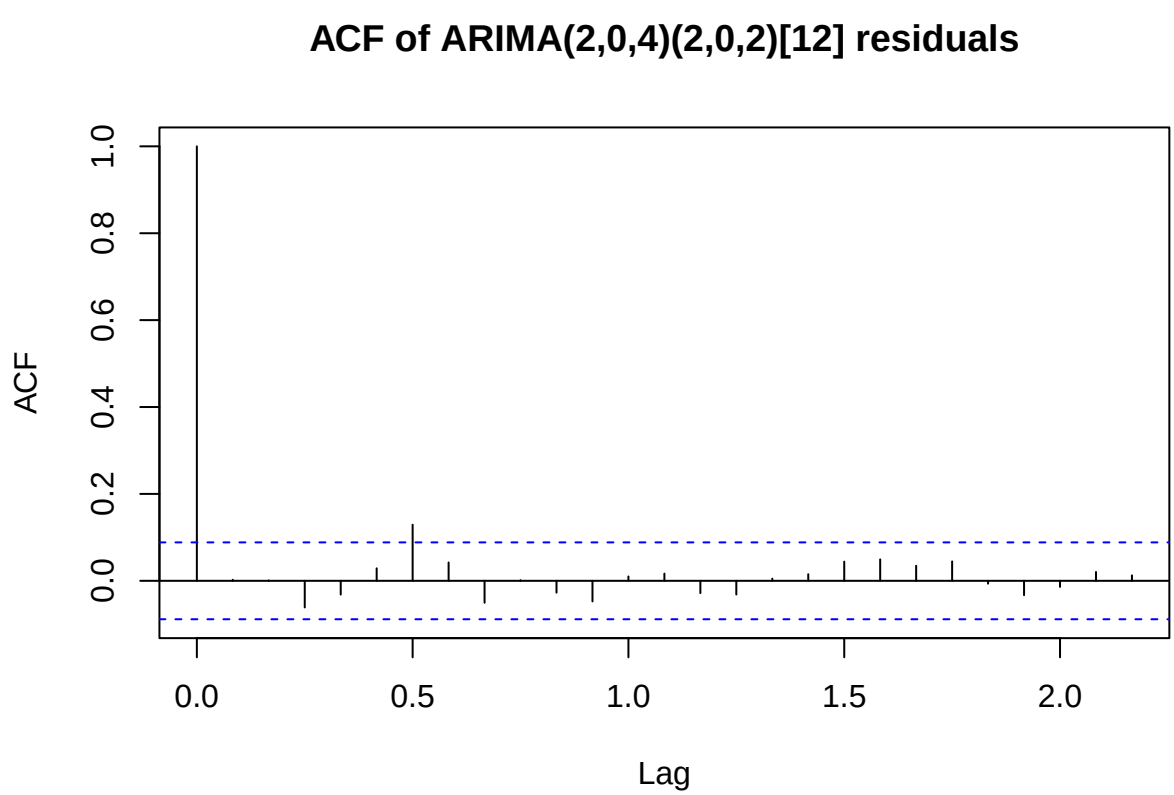
```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 4), seasonal = list(order = c(2,
```

```
##     0, 2), period = 12), method = "ML")
##
## Coefficients:

## Warning in sqrt(diag(x$var.coef)): NaNs produced

##           ar1     ar2     ma1      ma2      ma3     ma4    sar1     sar2
##       -0.0227  0.9456  0.2107  -0.8658  -0.0184  0.0582  1.1899  -0.3936
## s.e.   0.0263  0.0249  0.0524   0.0561   0.0507  0.0541     NaN      NaN
##          sma1    sma2  intercept
##       -1.2437  0.5093     0.0045
## s.e.      NaN     NaN     0.0015
##
## sigma^2 estimated as 2.392e-05:  log likelihood = 1914.05,  aic = -3804.1
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001308868 0.004891118 0.002897664 NaN  Inf 0.7657977
##                     ACF1
## Training set 0.002438932
```

```r
acf(residuals(test5), main = "ACF of ARIMA(2,0,4)(2,0,2)[12] residuals")
```



```r
# Comparing AIC values
aic = cbind(manual_fit$aic, test1$aic, test2$aic, test3$aic, test4$aic, test5$aic)
colnames(aic) = c("manual_fit", "test1", "test2", "test3", "test4", "test5")
aic
```

```
##      manual_fit     test1     test2     test3    test4     test5
## [1,]  -3803.702 -3801.817 -3807.156 -3806.146 -3801.32 -3804.098
```

```r
# Best (based on lowest AIC and ACF plot): test3, ARIMA(3,0,5)(2,0,1)[12]

order <- c(3,0,5)
seasonal <- list(order = c(2,0,1), period = 12)
# Fit the model to the training data
train_arima_model <- arima(montreal_train_ts, order = order,
                           seasonal = seasonal, method = "ML", include.mean = T)
```

```
## Warning in arima(montreal_train_ts, order = order, seasonal = seasonal, :
## possible convergence problem: optim gave code = 1
```

```r
# Result the moving 1-step ahead forecast for holdout set
arima_results <- arma_fc(tsdata = montreal_transformed_nhpi,
                         ntrain = length(montreal_train_ts),
                         order = order, seasonal = seasonal, method = "ML",
                         traincoef = coef(train_arima_model), include.mean = T)
arima_results$rmse
```

```
## [1] 0.008251754
```

Doing the ARIMAX forecasts with lagged interest

```r
# Method 3
montreal_fit_arimax <- forecast::auto.arima(montreal_train_ts, xreg = montreal_train_ir_xreg)
summary(montreal_fit_arimax)
```

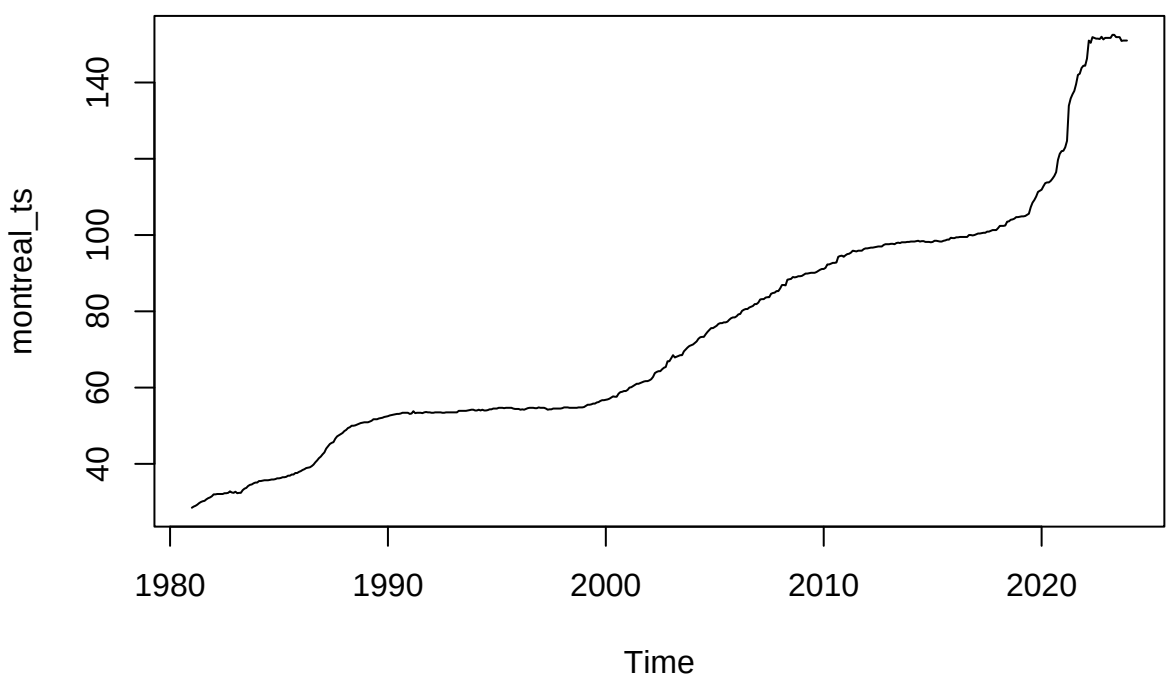
```
## Series: montreal_train_ts
## Regression with ARIMA(2,0,5)(2,0,1)[12] errors
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2998  -0.4778  -1.1411  0.3668  0.0793  -0.0121  0.1545  0.7148  0.0216
## s.e.  0.1554   0.1577   0.1572  0.1477  0.0684   0.0681  0.0629  0.2831  0.0728
##          sma1  intercept   xreg
##       -0.6608     0.0028  1e-04
## s.e.   0.2890     0.0010  1e-04
##
## sigma^2 = 2.435e-05:  log likelihood = 1916.72
## AIC=-3807.45   AICc=-3806.69   BIC=-3752.9
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE     MASE
```

```
## Training set -2.873012e-05 0.004873876 0.002945387 NaN  Inf 0.764944
##                      ACF1
## Training set 0.003074701
```

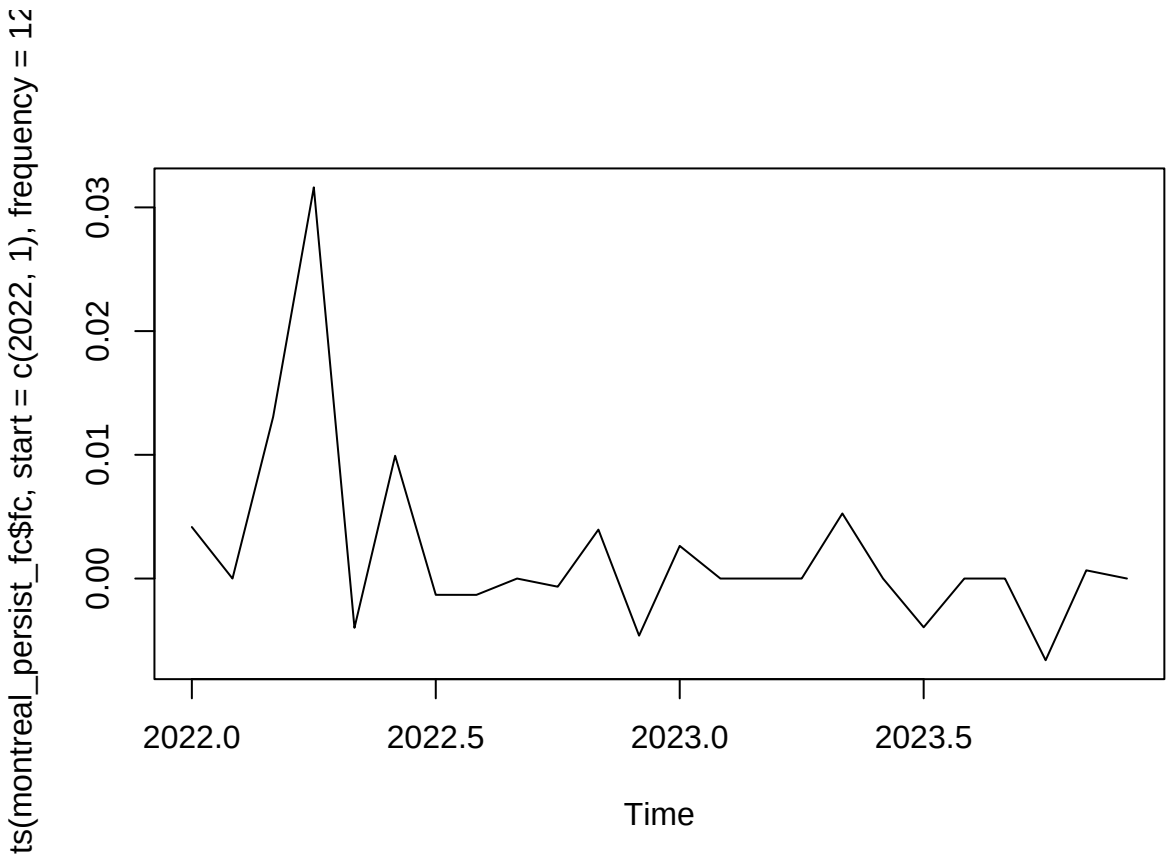
```
order <- c(3,0,5)
seasonal <- list(order = c(2,0,1), period = 12)
# Fitting a model to the training data
train_arimax_model <- arima(montreal_train_ts, order = order, seasonal = seasonal,
                            xreg = montreal_train_ir_xreg, method = "ML", include.mean = T)
# Getting the result of the moving 1-step ahead forecast for holdout set
arimax_results <- armax_fc(tsdata = montreal_transformed_nhpi,
                           ntrain = length(montreal_train_ts), order = order,
                           seasonal = seasonal, xreg = montreal_irlag1[-1],
                           method = "ML", traincoef = coef(train_arimax_model), include.mean = T)
```

Viewing the results of Montreal's analyses

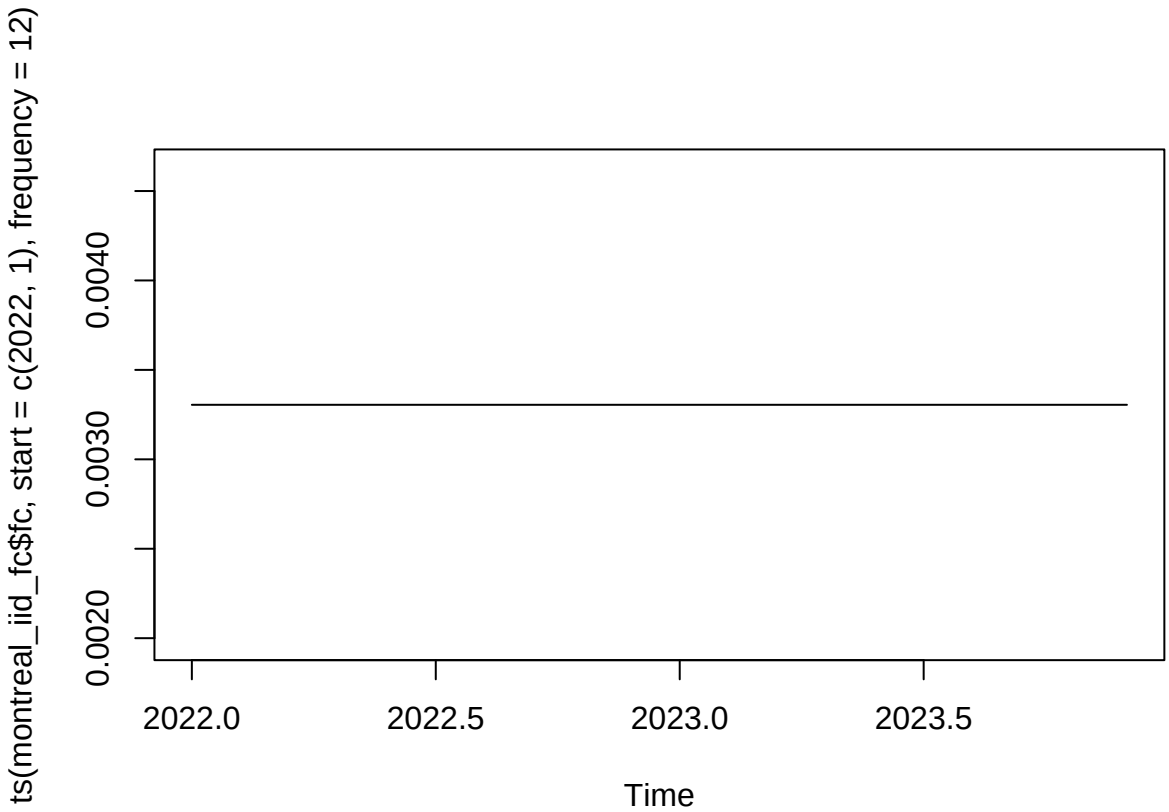
```
# 1. TIME SERIES PLOT
plot(montreal_ts)
```



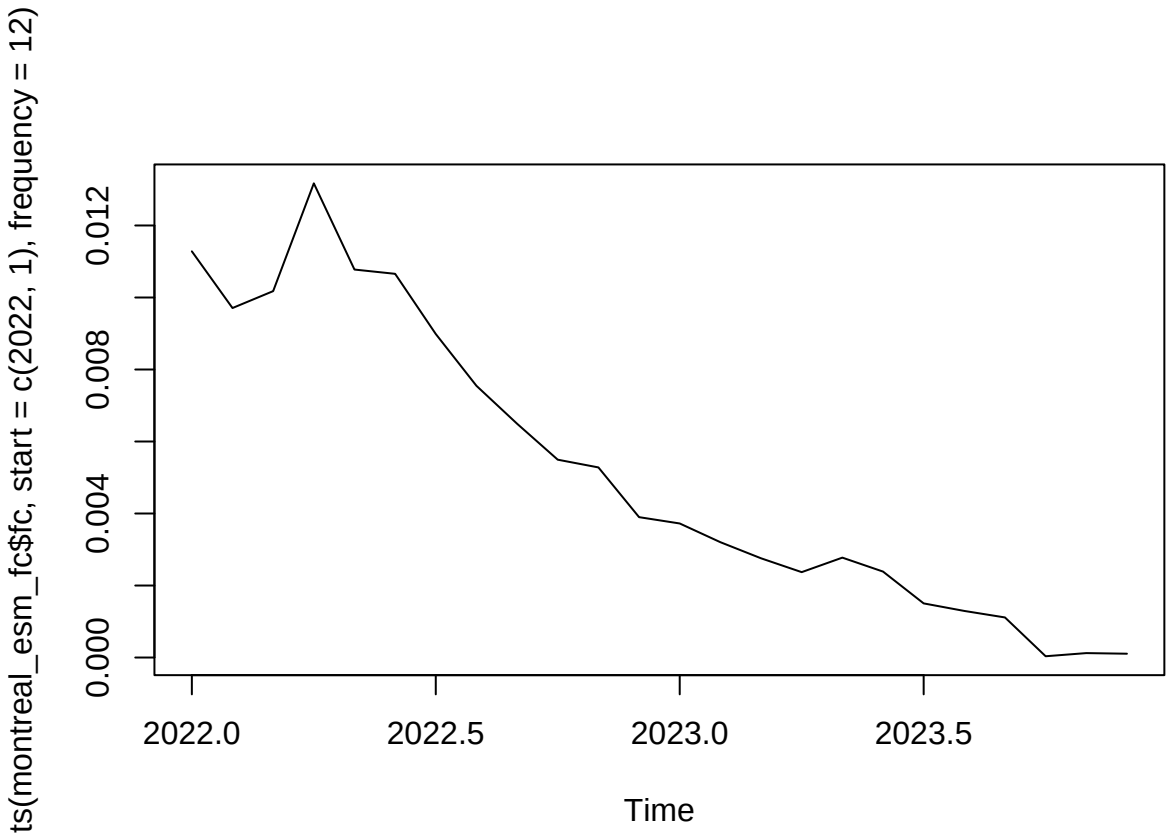

```
# log-diff transformation
# plot(montreal_transformed_ts) # NOTE: this object doesn't exist
# ts plot from persistence
plot(ts(montreal_persist_fc$fc, start = c(2022, 1), frequency = 12))
```



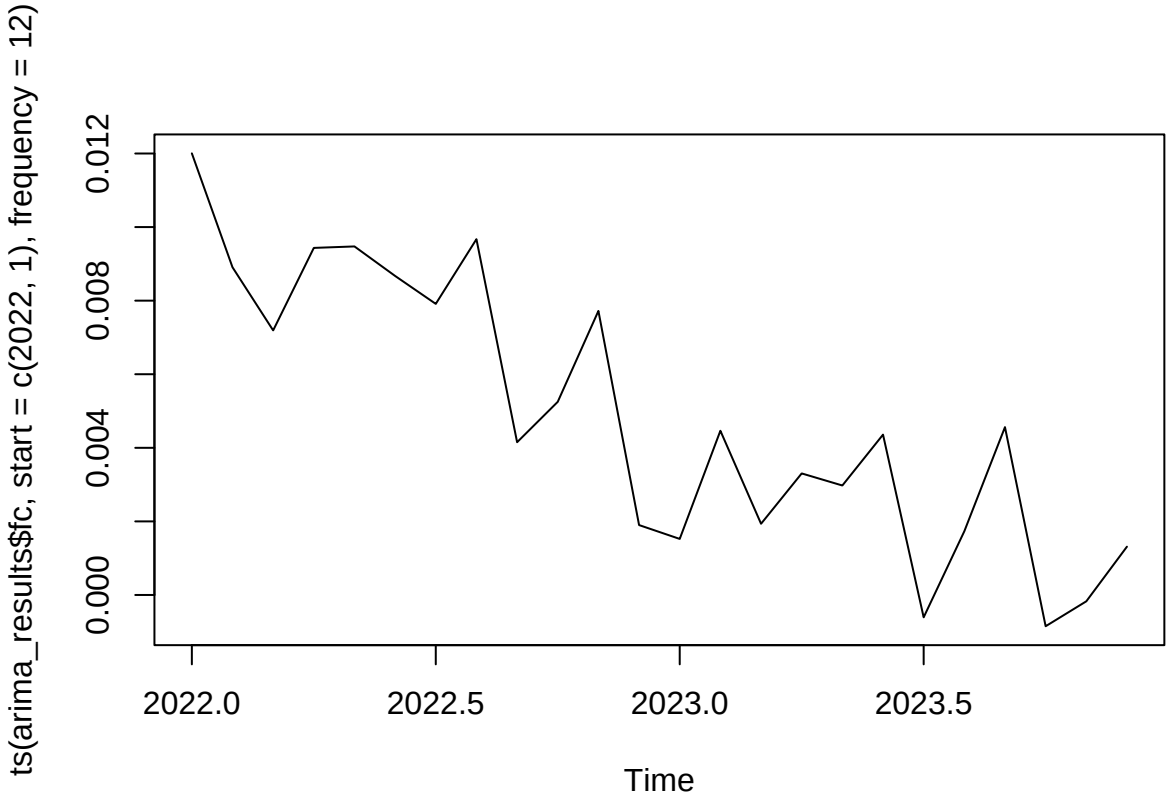

```r
# ts plot from average
plot(ts(montreal_iid_fc$fc, start = c(2022, 1), frequency = 12))
```



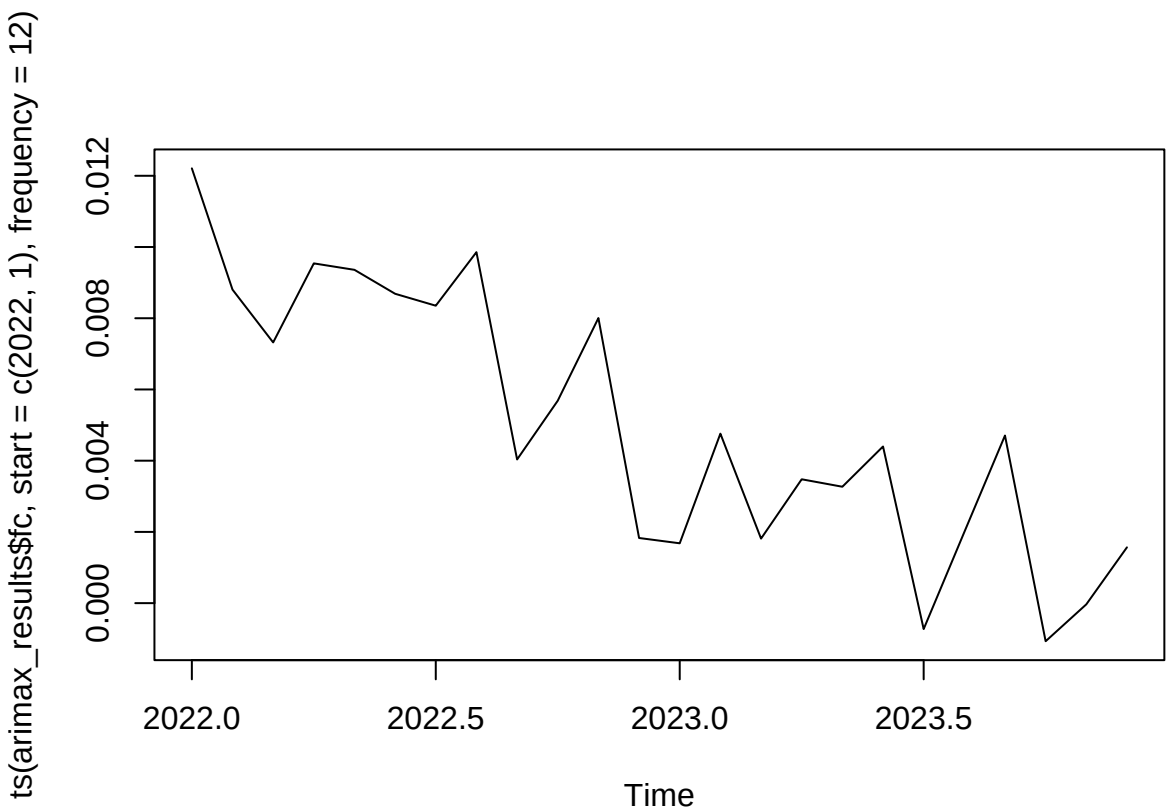

```r
# ts plot from simple exponential smoothing
plot(ts(montreal_esm_fc$fc, start = c(2022, 1), frequency = 12))
```



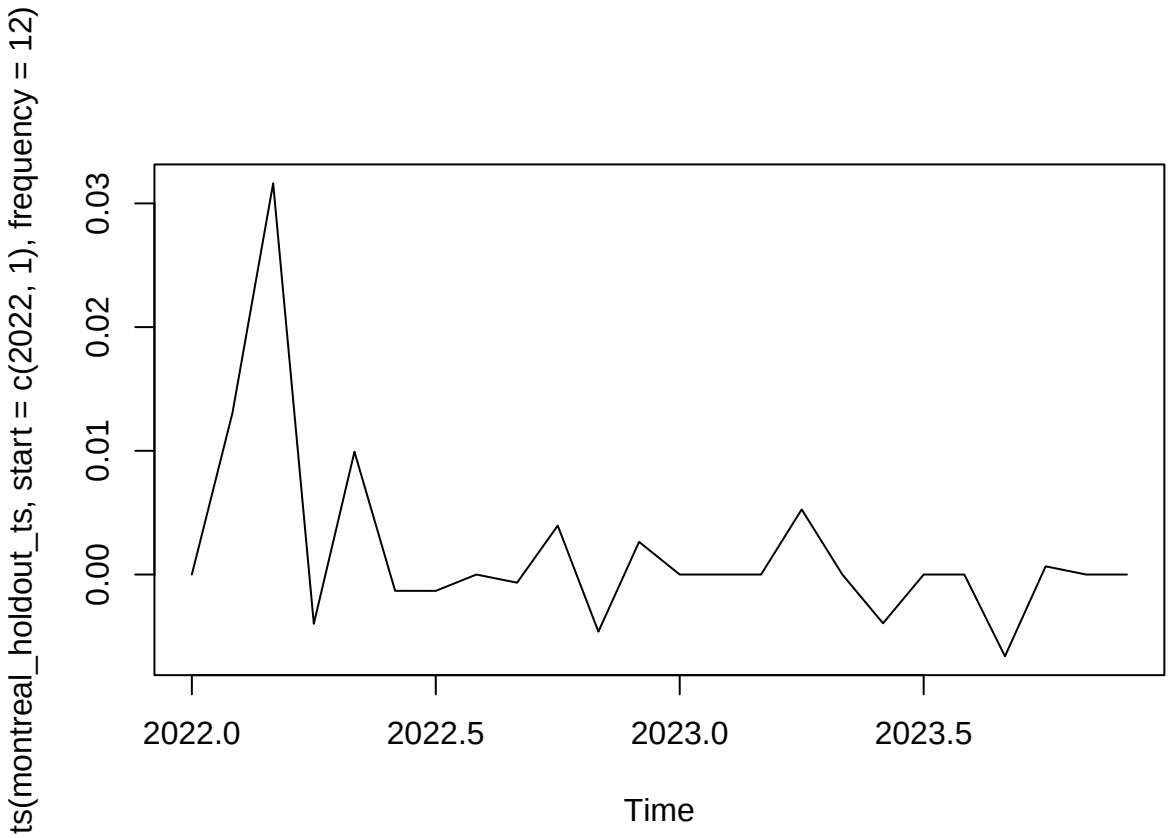

```r
# ts plot from arima
plot(ts(arima_results$fc, start = c(2022, 1), frequency = 12))
```



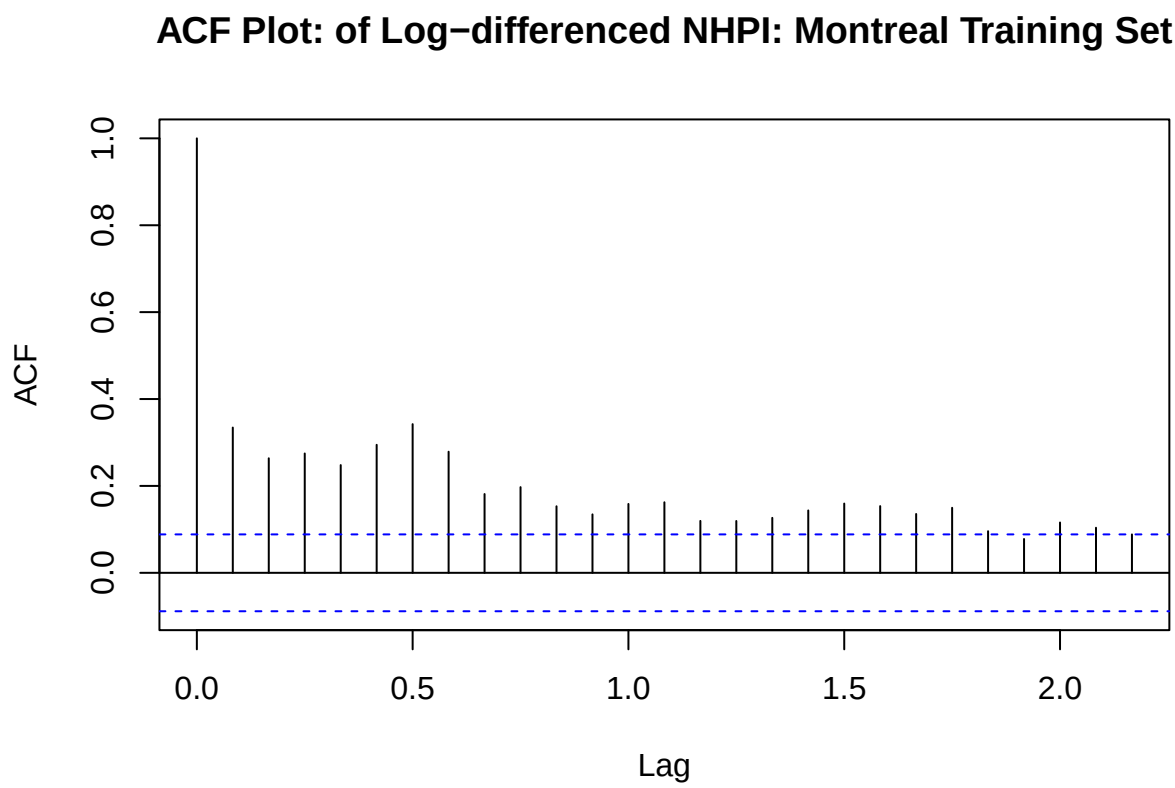

```r
# ts plot from arimax
plot(ts(arimax_results$fc, start = c(2022, 1), frequency = 12))
```



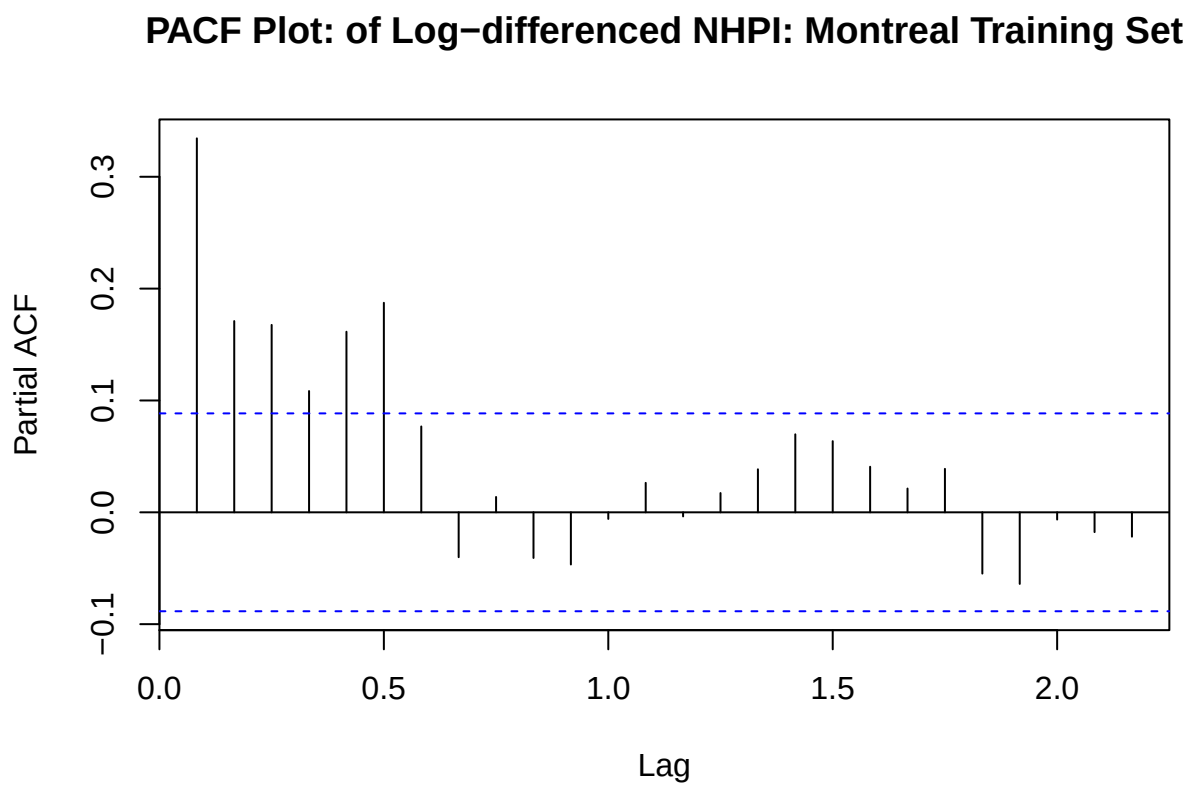

```r
# ts plot of holdout set
plot(ts(montreal_holdout_ts, start = c(2022, 1), frequency = 12))
```



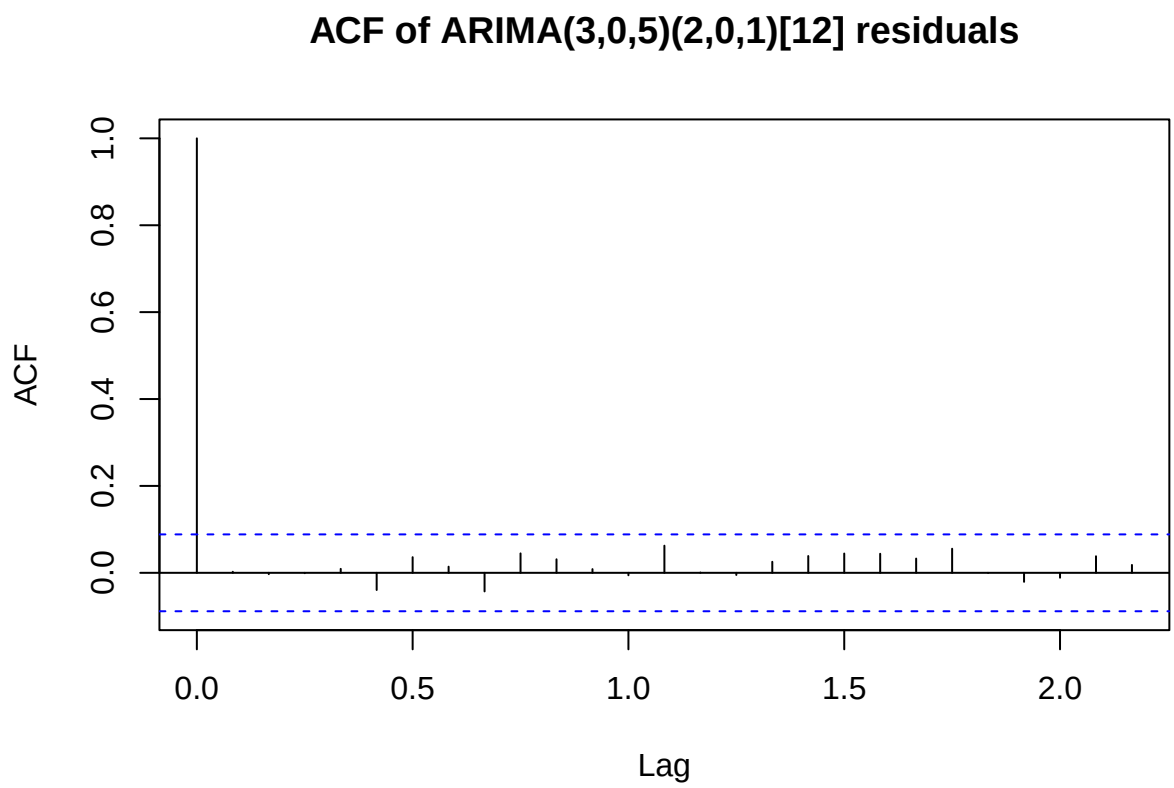

```r
# 2. ACF PLOT
# ACF and PACF of training set
acf(montreal_train_ts, main = "ACF Plot: of Log-differenced NHPI: Montreal Training Set")
```



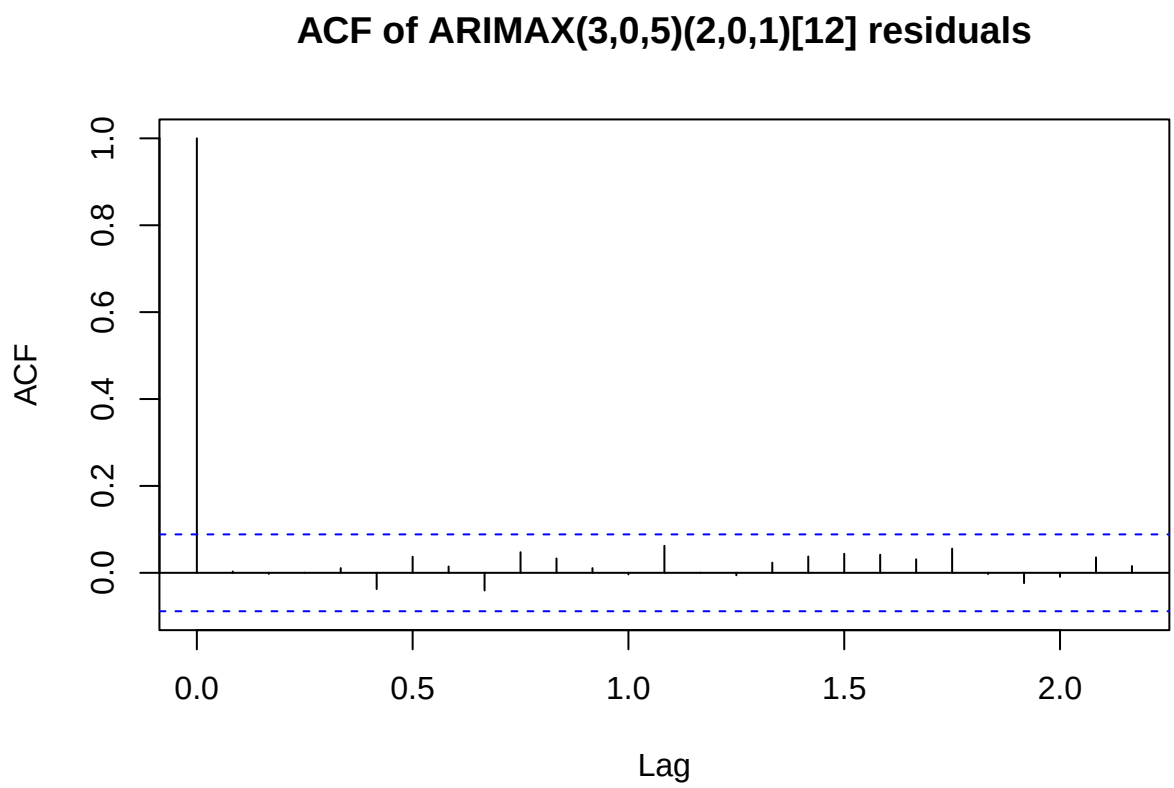

```r
pacf(montreal_train_ts, main = "PACF Plot: of Log-differenced NHPI: Montreal Training Set")
```




```r
# ACF from best ARIMA model
resid_fit <- residuals(montreal_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(3,0,5)(2,0,1)[12] residuals")
```




```r
# ACF from best ARIMAX model
resid_fit <- residuals(montreal_fit_arimax)
acf(resid_fit, main = "ACF of ARIMAX(3,0,5)(2,0,1)[12] residuals")
```




```r
# 3. SUMMARY STATISTICS
summary(montreal_train_ts)
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -0.012308  0.000000  0.001907  0.003305  0.005096  0.071182
```

```r
summary(montreal_holdout_ts)
```

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.0066050 -0.0008246  0.0000000  0.0018622  0.0011569  0.0316205
```

```r
# summary of best ARIMA model
summary(montreal_transformed_nphi_auto)
```

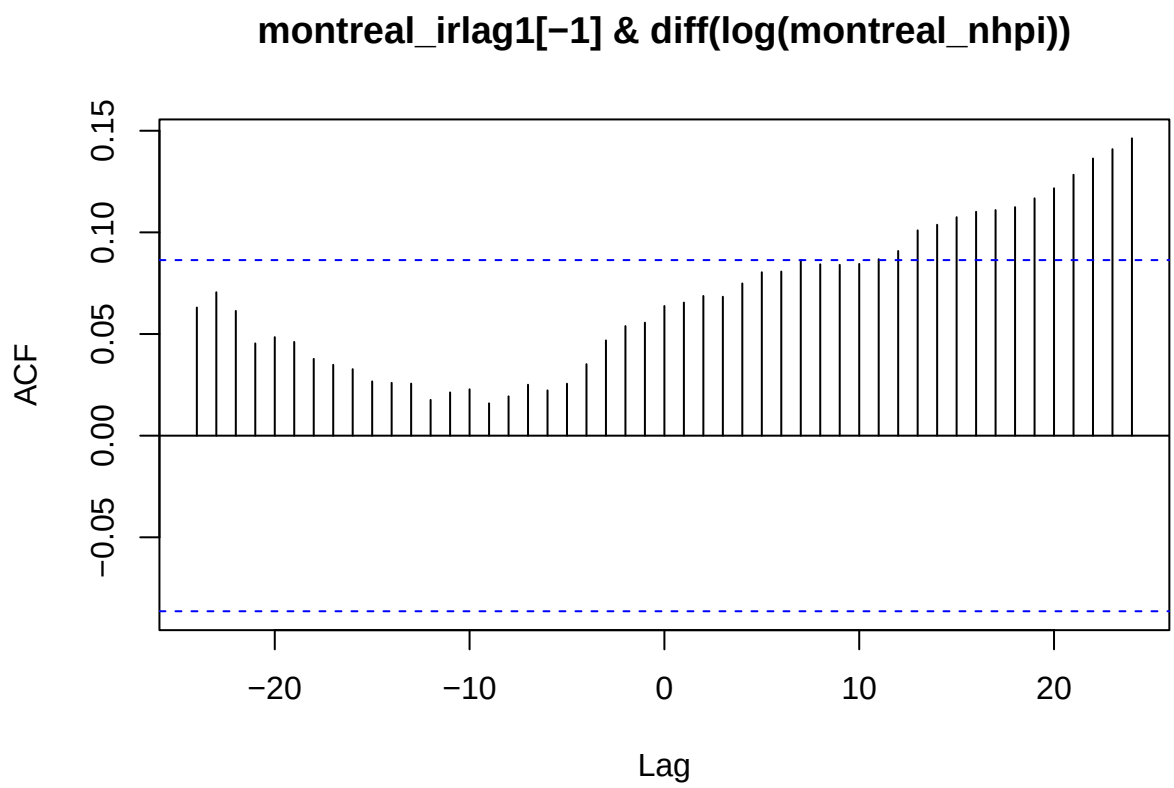
```
## Series: montreal_train_ts
## ARIMA(2,0,5)(2,0,1)[12] with non-zero mean
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2933  -0.4683  -1.1324  0.3576  0.0775  -0.0112  0.1526  0.7129  0.0270
## s.e.  0.1615   0.1643   0.1629  0.1522  0.0680   0.0678  0.0628  0.2571  0.0723
##          sma1    mean
##       -0.6604  0.0034
## s.e.   0.2622  0.0007
```

```
##
## sigma^2 = 2.434e-05:  log likelihood = 1916.27
## AIC=-3808.54   AICc=-3807.89   BIC=-3758.19
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -4.22165e-05 0.004878312 0.002960714 NaN  Inf 0.7689245
##                     ACF1
## Training set 0.002456102
```

```r
# summary of best ARIMAX model
summary(montreal_fit_arimax)
```

```
## Series: montreal_train_ts
## Regression with ARIMA(2,0,5)(2,0,1)[12] errors
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2998  -0.4778  -1.1411  0.3668  0.0793  -0.0121  0.1545  0.7148  0.0216
## s.e.  0.1554   0.1577   0.1572  0.1477  0.0684   0.0681  0.0629  0.2831  0.0728
##          sma1  intercept   xreg
##       -0.6608     0.0028  1e-04
## s.e.   0.2890     0.0010  1e-04
##
## sigma^2 = 2.435e-05:  log likelihood = 1916.72
## AIC=-3807.45   AICc=-3806.69   BIC=-3752.9
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE     MASE
## Training set -2.873012e-05 0.004873876 0.002945387 NaN  Inf 0.764944
##                     ACF1
## Training set 0.003074701
```

```r
# CCF between log-differenced NHPI and the lagged interest rate
ccf(montreal_irlag1[-1], diff(log(montreal_nhpi)))
```




```r
# 4. RMSE
rmse = cbind(montreal_persist_fc$rmse, montreal_iid_fc$rmse,
             montreal_esm_fc$rmse, arima_results$rmse, arimax_results$rmse)
colnames(rmse) = c("persist", "iid", "esm", "arima", "arimax")
print(round(rmse, 3))
```

```
##   persist   iid   esm arima arimax
## a    0.01 0.008 0.008 0.008  0.008
```

Analyzing Toronto

Setting up

```r
# Reading the data
toronto_nhpi_data <- read.csv("Full_Cities/toronto_full.csv")
toronto_nhpi_ir_data <- read.csv("Extracted_Cities/toronto.csv")

# Extracting the NHPI
toronto_nhpi <- toronto_nhpi_data$VALUE

# Extracting the one-month lag interest rate
toronto_irlag1 <- toronto_nhpi_ir_data$IR_lag1
```

```r
# Setting the time window from 1981-2023
toronto_ts <- ts(toronto_nhpi, start = c(1981,1), end = c(2023, 12), frequency = 12)
toronto_ir_ts <- ts(toronto_irlag1, start = c(1981,1), end = c(2023, 12), frequency = 12)

# Applying a transformation for NHPI
toronto_transformed_nhpi <- diff(log(toronto_ts))

# Setting a training set from 1981-2021
toronto_train_ts <- window(toronto_transformed_nhpi, end = c(2021,12))
toronto_train_ir_xreg <- window(toronto_ir_ts, start = c(1981,2), end = c(2021,12))

# Setting a holdout set from 2022-2023
toronto_holdout_ts <- window(toronto_transformed_nhpi, start = c(2022,1))
toronto_holdout_ir_xreg <- window(toronto_ir_ts, start = c(2022,1))
```

Doing the baseline forecasts

```r
# Baseline 1
toronto_persist_fc <- persist_fc(toronto_train_ts, toronto_holdout_ts, iprint = F)

# Baseline 2
toronto_iid_fc <- iid_fc(toronto_train_ts, toronto_holdout_ts, iprint = F)
```

Doing the exponential smoothing forecasts

```r
# Method 1
ntrain <- length(toronto_train_ts)
ntotal <- length(toronto_ts)-1

toronto_transformed_nhpi_train <- toronto_transformed_nhpi[1:ntrain]
toronto_transformed_nhpi_holdout <- toronto_transformed_nhpi[(ntrain+1):ntotal]

# Simple Exponential Smoothing
fit_expsmo <- HoltWinters(toronto_train_ts, beta = F, gamma = F)
toronto_esm_fc <- esm_fc(toronto_transformed_nhpi_train, toronto_transformed_nhpi_holdout,
                         alpha = fit_expsmo$alpha, level = fit_expsmo$coefficients["a"],
                         iprint = F)

# Linear Exponential Smoothing
fit_hw <- HoltWinters(toronto_train_ts, gamma = F)
toronto_holt_fc <- lholt_fc(toronto_transformed_nhpi_train, toronto_transformed_nhpi_holdout,
                            alpha = fit_hw$alpha, beta = fit_hw$beta,
                            level = fit_hw$coefficients["a"],
                            slope = fit_hw$coefficients["b"], iprint = F)

# Additive Seasonal Exponential Smoothing
wafit <- HoltWinters(toronto_train_ts, seasonal = "additive")
toronto_aseason_fc <- aseason_fc(toronto_transformed_nhpi_train,
                                 toronto_transformed_nhpi_holdout,
                                 alpha = wafit$alpha, beta = wafit$beta,
```

```r
                                gamma = wafit$gamma, level = wafit$coef[1],
                                slope = wafit$coef[2], season = wafit$coefficient[3:14],
                                iprint = F)

# Comparing RMSE
rmse_esm <- cbind(toronto_esm_fc$rmse, toronto_holt_fc$rmse, toronto_aseason_fc$rmse)
colnames(rmse_esm) <- c("esm", "lholt", "aseason")
rmse_esm
```

```
##           esm       lholt     aseason
## a 0.002166508 0.002334658 0.002612987
```

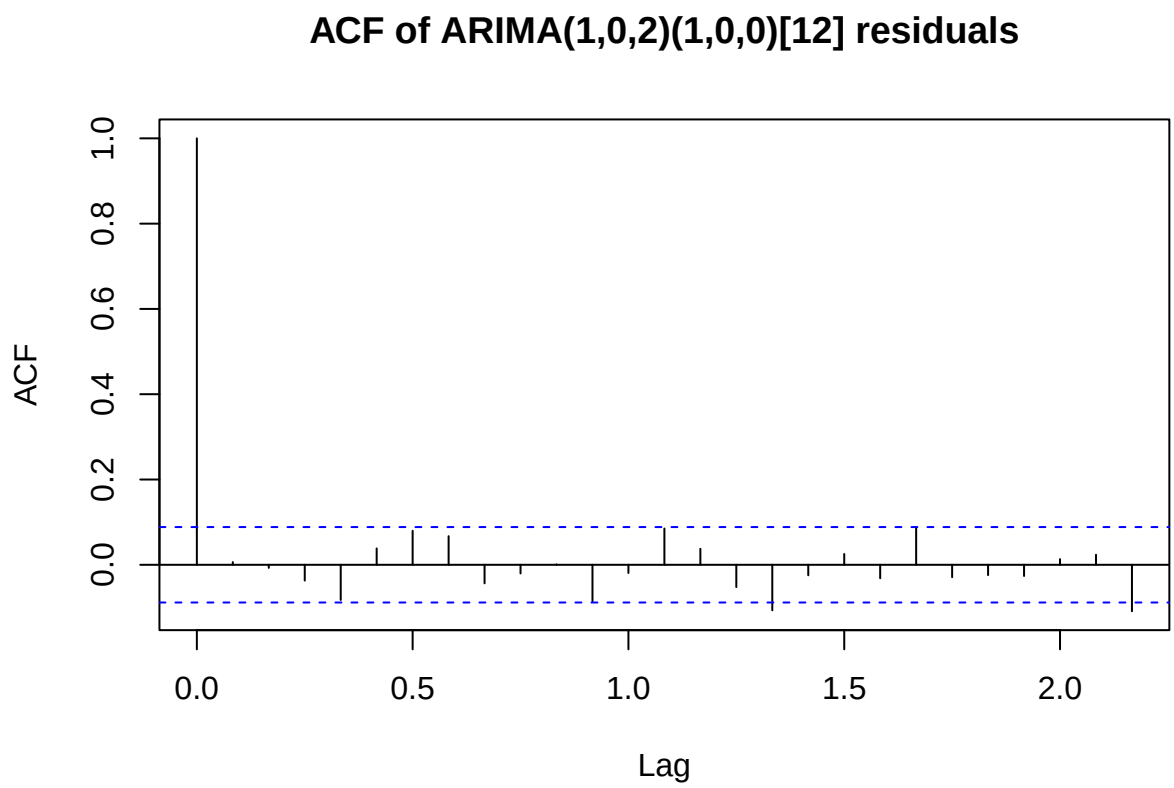
```r
# Best exponential smoothing method: Simple Exponential Smoothing
```

Doing the ARIMA forecasts

```r
# Method 2
toronto_transformed_nphi_auto <- forecast::auto.arima(toronto_train_ts)
summary(toronto_transformed_nphi_auto) # 0.0064057
```

```
## Series: toronto_train_ts
## ARIMA(1,0,2)(1,0,0)[12] with non-zero mean
##
## Coefficients:
##          ar1      ma1      ma2    sar1    mean
##       0.9161  -0.4317  -0.1386  0.1316  0.0034
## s.e.  0.0302   0.0551   0.0490  0.0487  0.0017
##
## sigma^2 = 4.146e-05:  log likelihood = 1782.61
## AIC=-3553.23   AICc=-3553.05   BIC=-3528.05
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001299236 0.006405744 0.003868432 NaN  Inf 0.6292045
##                     ACF1
## Training set 0.006342322
```

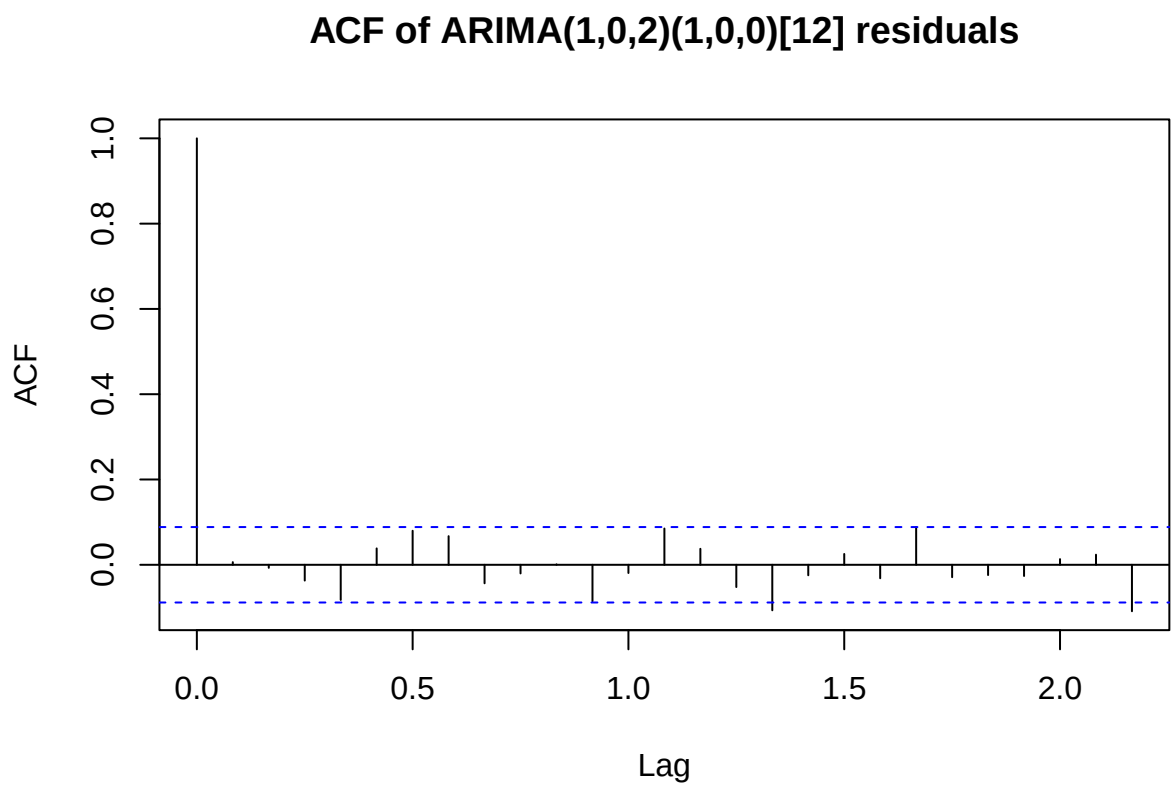
```r
resid_fit <- residuals(toronto_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(1,0,2)(1,0,0)[12] residuals")
```




```r
# Comparing with the other ARIMA models
manual_fit <- arima(toronto_train_ts, order = c(1,0,2),
                    seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(manual_fit) # RMSE: 0.006405747
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 2), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1      ma2    sar1  intercept
##       0.9160  -0.4315  -0.1388  0.1316     0.0034
## s.e.  0.0302   0.0551   0.0490  0.0487     0.0017
##
## sigma^2 estimated as 4.103e-05:  log likelihood = 1782.61,  aic = -3553.23
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001276383 0.006405747 0.003868164 NaN  Inf 0.8546019
##                     ACF1
## Training set 0.006235644
```

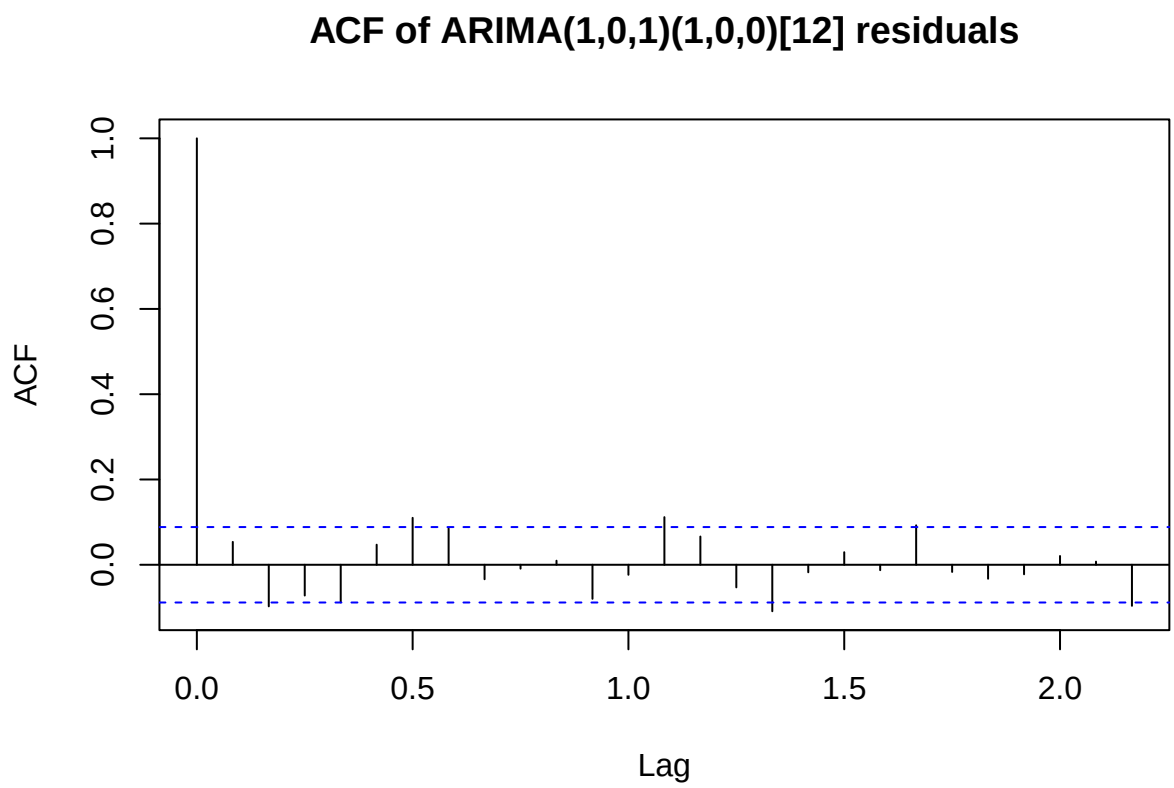
```r
acf(residuals(manual_fit), main = "ACF of ARIMA(1,0,2)(1,0,0)[12] residuals")
```




```r
test1 <- arima(toronto_train_ts, order = c(1,0,1),
               seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(test1) # RMSE: 0.006456026
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 1), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1    sar1  intercept
##       0.8636  -0.4245  0.1466     0.0033
## s.e.  0.0434   0.0829  0.0482     0.0014
##
## sigma^2 estimated as 4.168e-05:  log likelihood = 1778.8,  aic = -3547.61
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001074825 0.006456026 0.003832997 NaN  Inf 0.8468326
##                    ACF1
## Training set 0.05363762
```

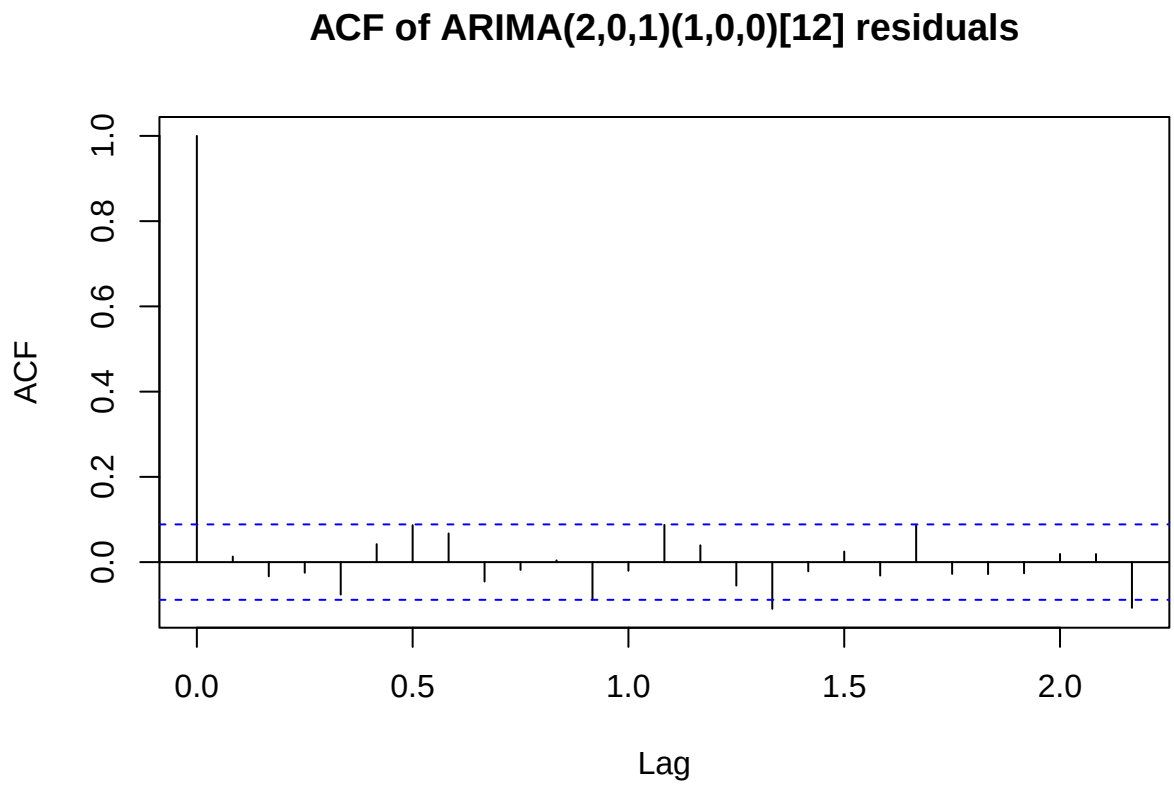
```r
acf(residuals(test1), main = "ACF of ARIMA(1,0,1)(1,0,0)[12] residuals")
```




```r
test2 <- arima(toronto_train_ts, order = c(2,0,1),
               seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(test2) # RMSE: 0.006410017
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(2, 0, 1), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2      ma1    sar1  intercept
##       1.1623  -0.2244  -0.6839  0.1292     0.0034
## s.e.  0.0959   0.0782   0.0792  0.0489     0.0017
##
## sigma^2 estimated as 4.109e-05:  log likelihood = 1782.3,  aic = -3552.6
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE       ACF1
## Training set -0.000128604 0.006410017 0.003868191 NaN  Inf 0.8546079 0.01298083
```

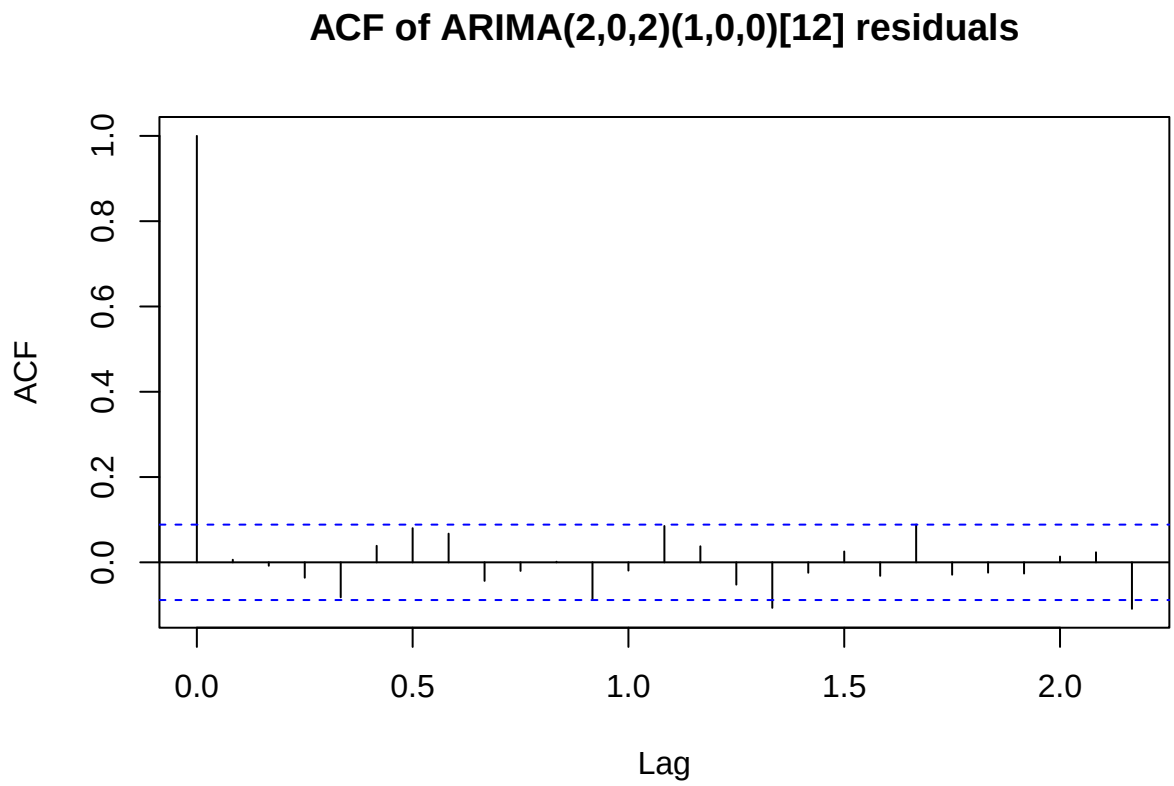
```r
acf(residuals(test2), main = "ACF of ARIMA(2,0,1)(1,0,0)[12] residuals")
```




```r
test3 <- arima(toronto_train_ts, order = c(2,0,2),
               seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(test3) # RMSE: 0.006405754
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(2, 0, 2), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2      ma1      ma2    sar1  intercept
##       0.9255  -0.0089  -0.4408  -0.1336  0.1316     0.0035
## s.e.  0.3544   0.3170   0.3512   0.1762  0.0491     0.0017
##
## sigma^2 estimated as 4.103e-05:  log likelihood = 1782.61,  aic = -3551.23
##
## Training set error measures:
##                        ME        RMSE        MAE MPE MAPE      MASE
## Training set -0.0001365834 0.006405754 0.00386938 NaN  Inf 0.8548707
##                     ACF1
## Training set 0.006071536
```

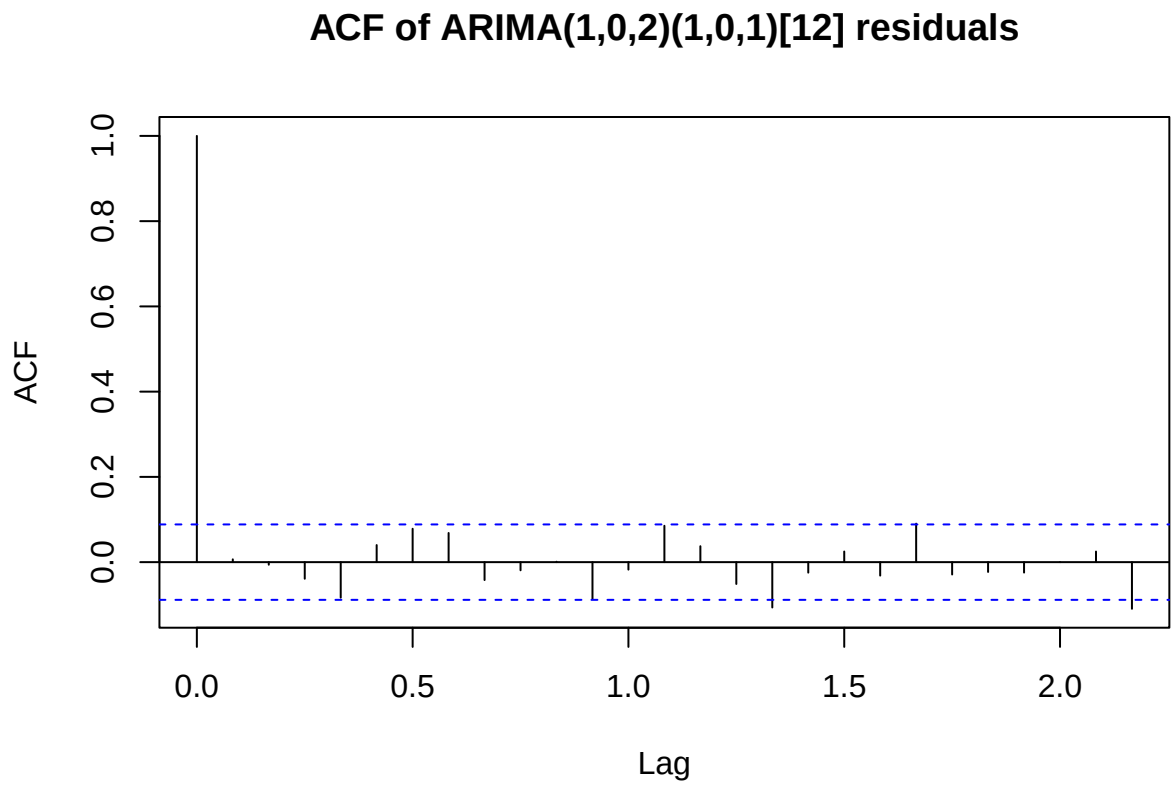
```r
acf(residuals(test3), main = "ACF of ARIMA(2,0,2)(1,0,0)[12] residuals")
```




```r
test4 <- arima(toronto_train_ts, order = c(1,0,2),
               seasonal = list(order = c(1,0,1), period = 12), method = "ML")
summary(test4) # RMSE: 0.006404998
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 2), seasonal = list(order = c(1,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1      ma2    sar1     sma1  intercept
##       0.9160  -0.4317  -0.1376  0.2437  -0.1140     0.0035
## s.e.  0.0303   0.0551   0.0491  0.3689   0.3775     0.0017
##
## sigma^2 estimated as 4.102e-05:  log likelihood = 1782.67,  aic = -3551.33
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001376316 0.006404998 0.003866659 NaN  Inf 0.8542695
##                     ACF1
## Training set 0.006611957
```

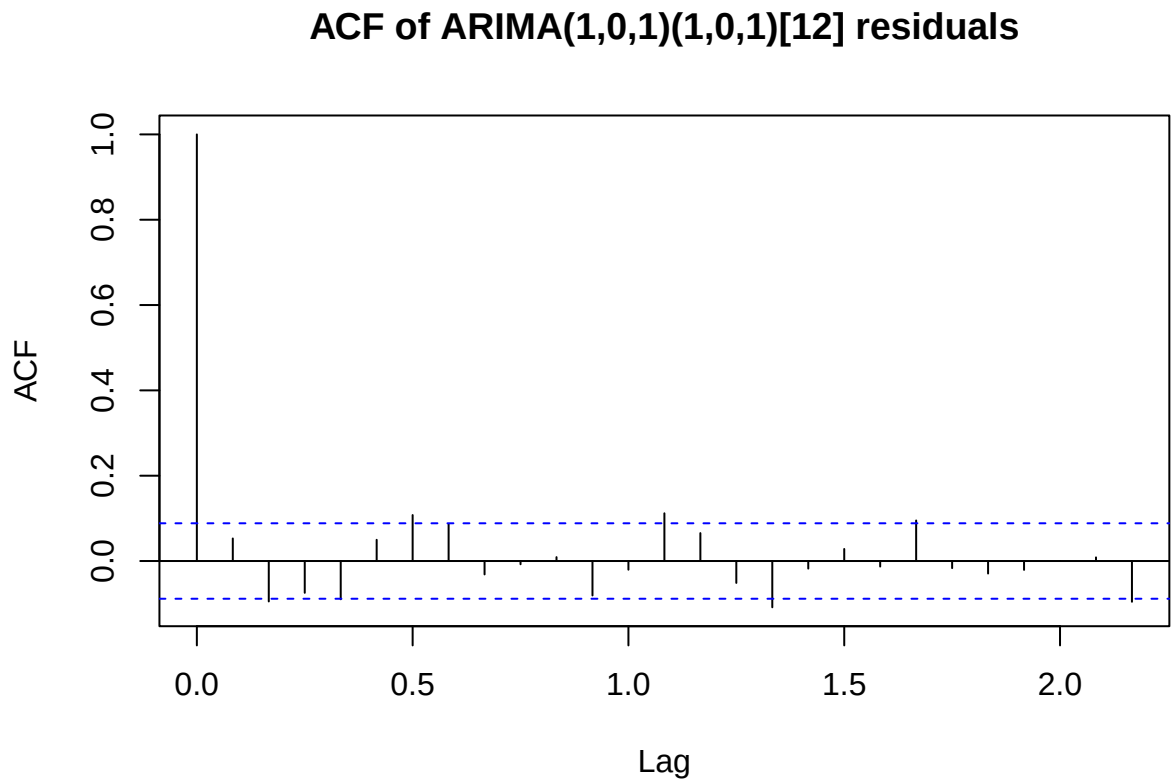
```r
acf(residuals(test4), main = "ACF of ARIMA(1,0,2)(1,0,1)[12] residuals")
```




```r
test5 <- arima(toronto_train_ts, order = c(1,0,1),
               seasonal = list(order = c(1,0,1), period = 12), method = "ML")
summary(test5) # RMSE: 0.006454211
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 1), seasonal = list(order = c(1,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1    sar1     sma1  intercept
##       0.8634  -0.4227  0.3043  -0.1614     0.0033
## s.e.  0.0434   0.0830  0.2853   0.2956     0.0015
##
## sigma^2 estimated as 4.166e-05:  log likelihood = 1778.93,  aic = -3545.87
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001083458 0.006454211 0.003830568 NaN  Inf 0.8462959
##                     ACF1
## Training set 0.05289892
```

```r
acf(residuals(test5), main = "ACF of ARIMA(1,0,1)(1,0,1)[12] residuals")
```




```r
# Comparing AIC values
aic = cbind(manual_fit$aic, test1$aic, test2$aic, test3$aic, test4$aic, test5$aic)
colnames(aic) = c("manual_fit", "test1", "test2", "test3", "test4", "test5")
aic
```

```
##      manual_fit     test1     test2    test3     test4     test5
## [1,]  -3553.228 -3547.608 -3552.598 -3551.23 -3551.334 -3545.868
```

```r
# Best (based on lowest AIC and ACF plot): manual_fit, ARIMA(1,0,2)(1,0,0)[12]

order <- c(1,0,2)
seasonal <- list(order = c(1,0,0), period = 12)
# Fitting the model to the training data
train_arima_model <- arima(toronto_train_ts, order = order,
                           seasonal = seasonal, method = "ML", include.mean = T)
# Getting the result of the moving 1-step ahead forecast for holdout set
arima_results <- arma_fc(tsdata = toronto_transformed_nhpi,
                         ntrain = length(toronto_train_ts), order = order,
                         seasonal = seasonal, method = "ML",
                         traincoef = coef(train_arima_model), include.mean = T)
arima_results$rmse
```

```
## [1] 0.00204875
```

Doing the ARIMAX forecasts with lagged interest

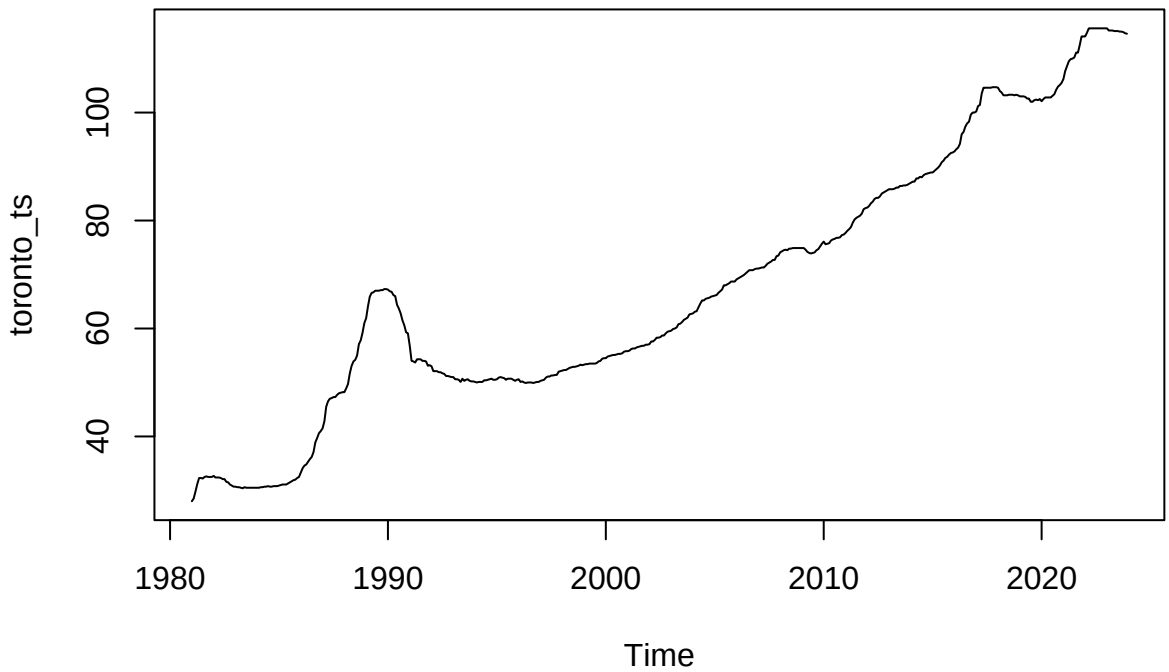
```
# Method 3
toronto_fit_arimax <- forecast::auto.arima(toronto_train_ts, xreg = toronto_train_ir_xreg)
summary(toronto_fit_arimax)
```

```
## Series: toronto_train_ts
## Regression with ARIMA(1,0,2)(1,0,0)[12] errors
##
## Coefficients:
##          ar1      ma1      ma2    sar1  intercept    xreg
##       0.9257  -0.4415  -0.1406  0.1330     0.0051  -3e-04
## s.e.  0.0329   0.0563   0.0488  0.0489     0.0028   4e-04
##
## sigma^2 = 4.147e-05:  log likelihood = 1783
## AIC=-3551.99   AICc=-3551.76   BIC=-3522.62
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001599343 0.006400238 0.003875959 NaN  Inf 0.6304288
##                     ACF1
## Training set 0.006084236
```

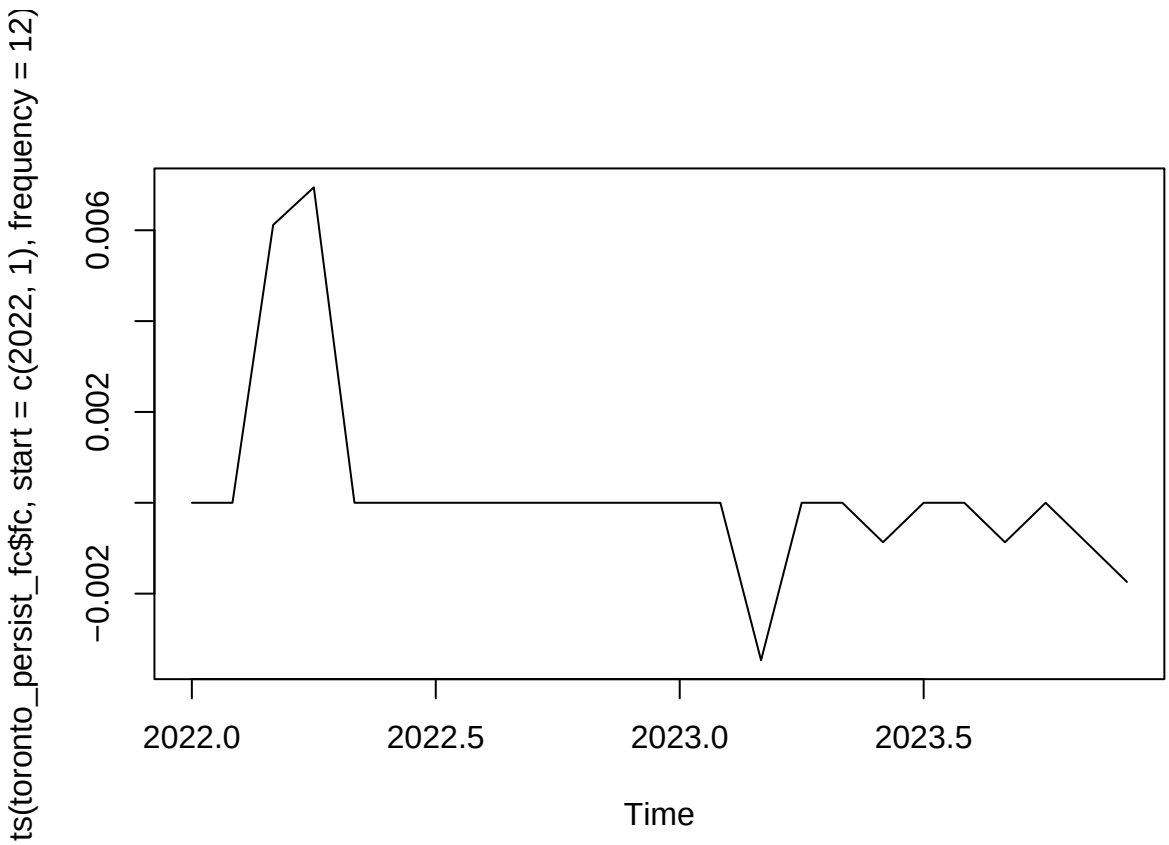
```
order <- c(1,0,2)
seasonal <- list(order = c(1,0,0), period = 12)
# Fitting a model to the training data
train_arimax_model <- arima(toronto_train_ts, order = order, seasonal = seasonal, xreg = toronto_train_
# Getting the result of the moving 1-step ahead forecast for holdout set
arimax_results <- armax_fc(tsdata = toronto_transformed_nhpi, ntrain = length(toronto_train_ts), order =
```

Viewing the results of Toronto's analyses

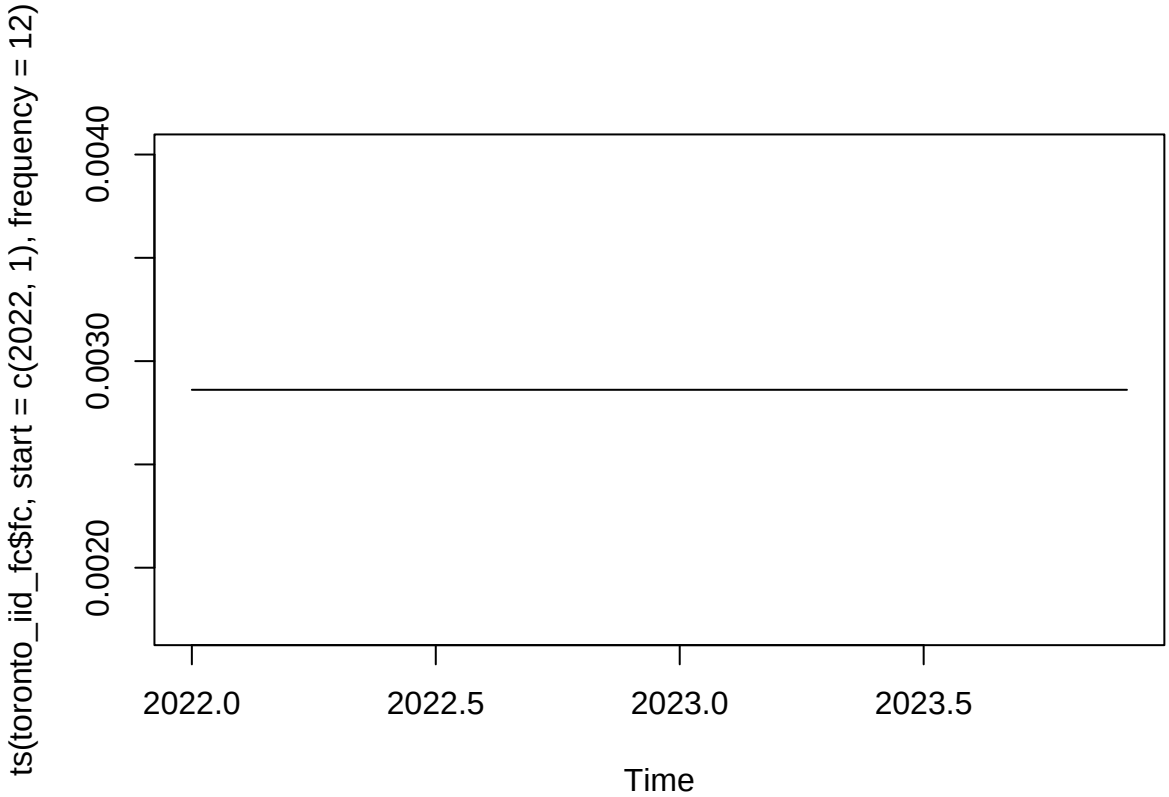
```
# 1. TIME SERIES PLOT
plot(toronto_ts)
```



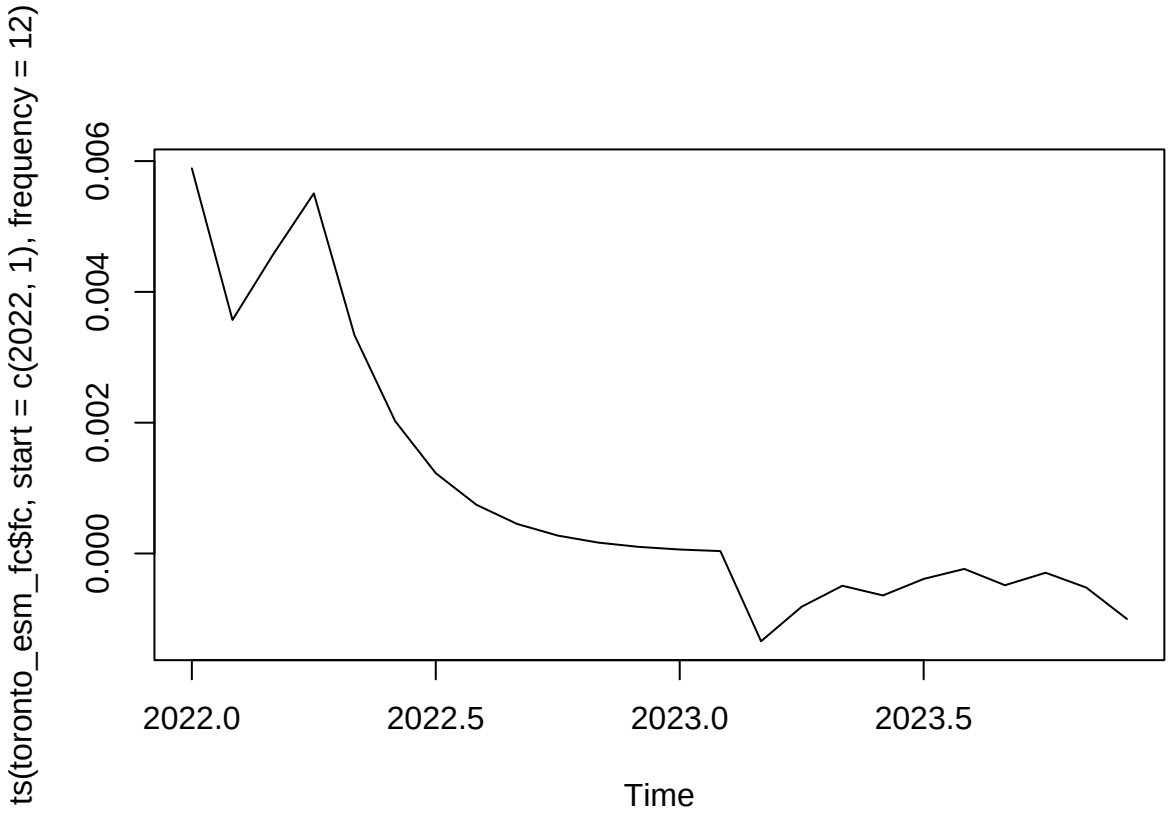

```
# log-diff transformation
# plot(toronto_transformed_ts) # NOTE: this object doesn't exist
# ts plot from persistence
plot(ts(toronto_persist_fc$fc, start = c(2022, 1), frequency = 12))
```



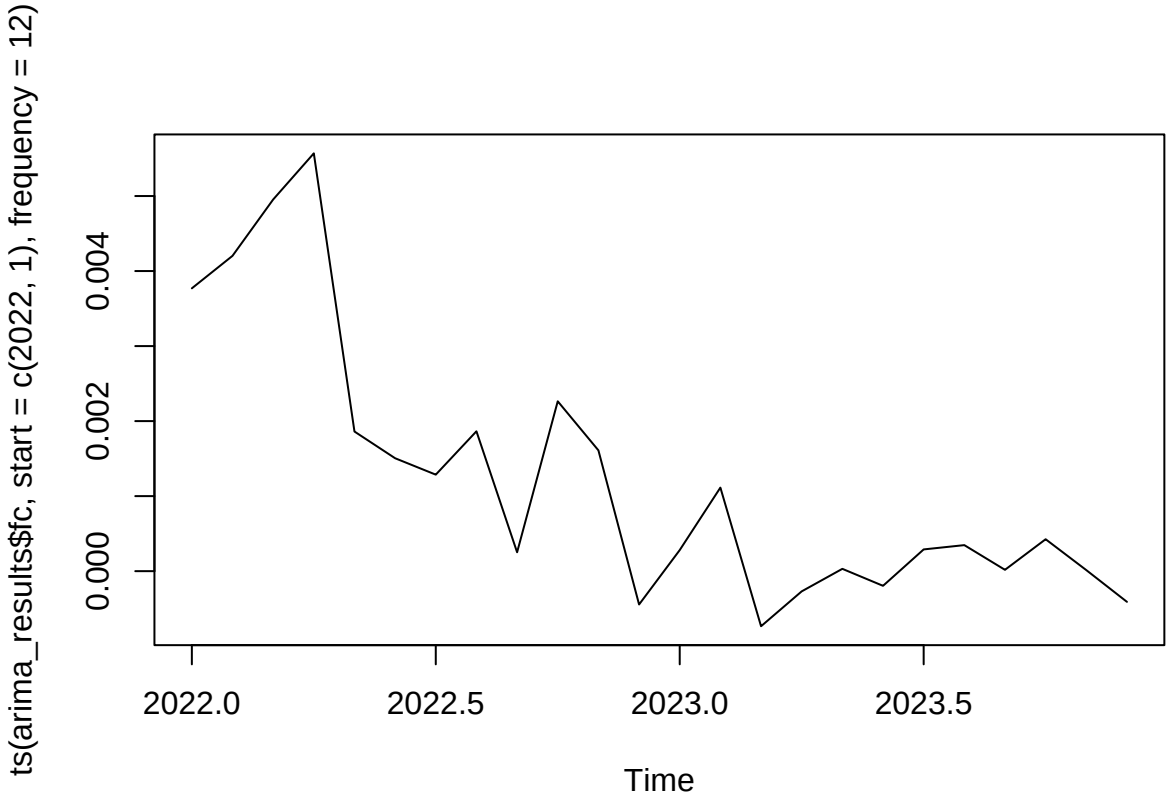

```r
# ts plot from average
plot(ts(toronto_iid_fc$fc, start = c(2022, 1), frequency = 12))
```



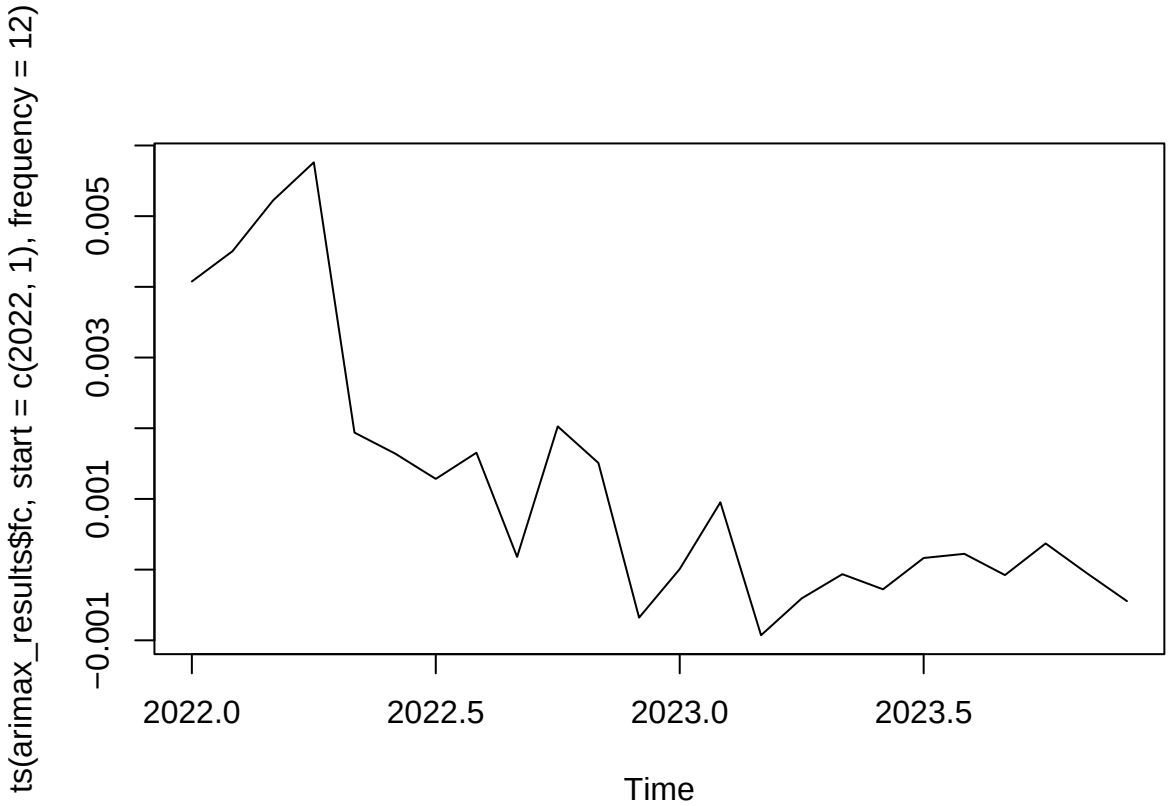

```r
# ts plot from simple exponential smoothing
plot(ts(toronto_esm_fc$fc, start = c(2022, 1), frequency = 12))
```



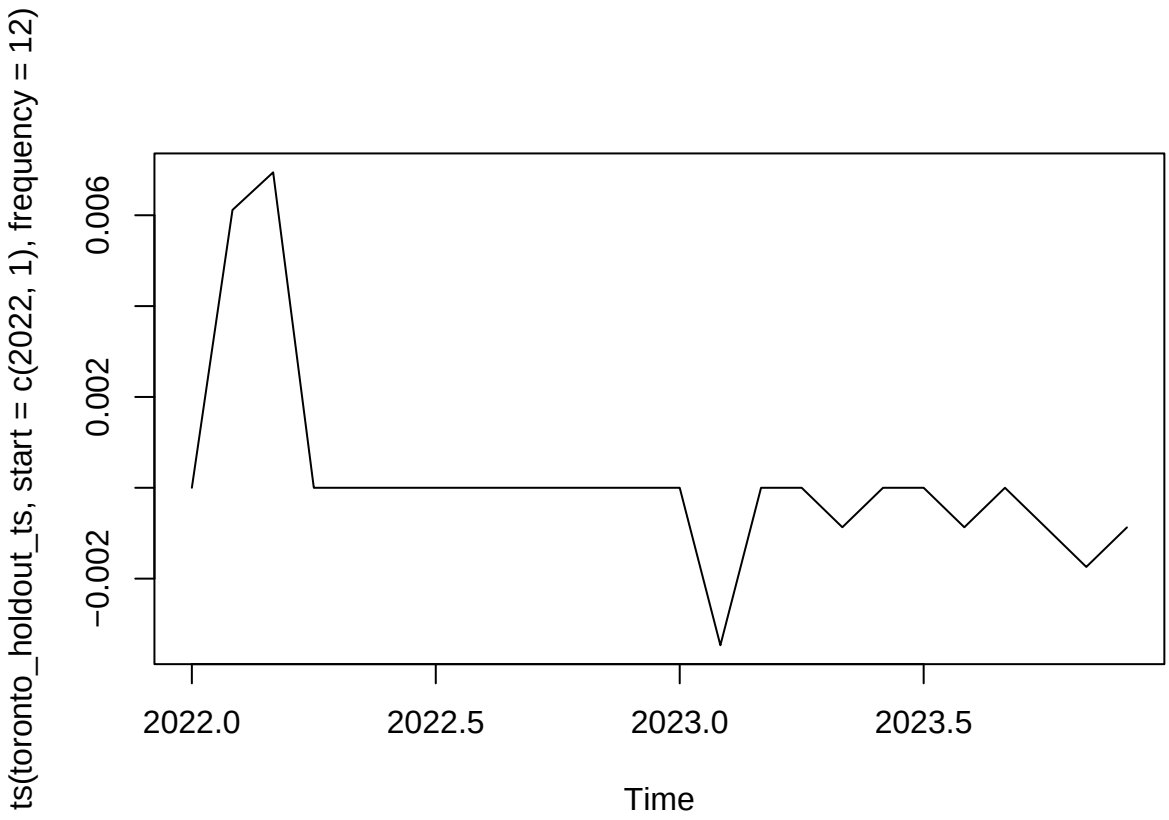

```r
# ts plot from arima
plot(ts(arima_results$fc, start = c(2022, 1), frequency = 12))
```



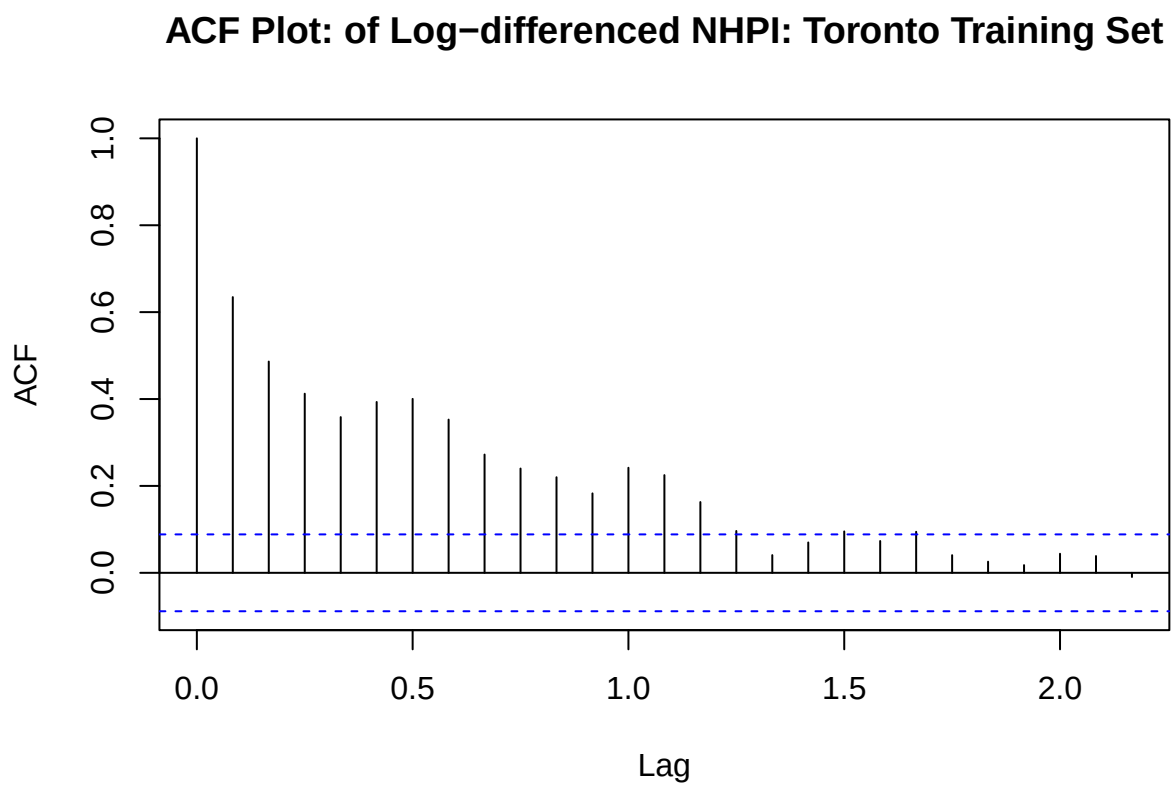

```r
# ts plot from arimax
plot(ts(arimax_results$fc, start = c(2022, 1), frequency = 12))
```



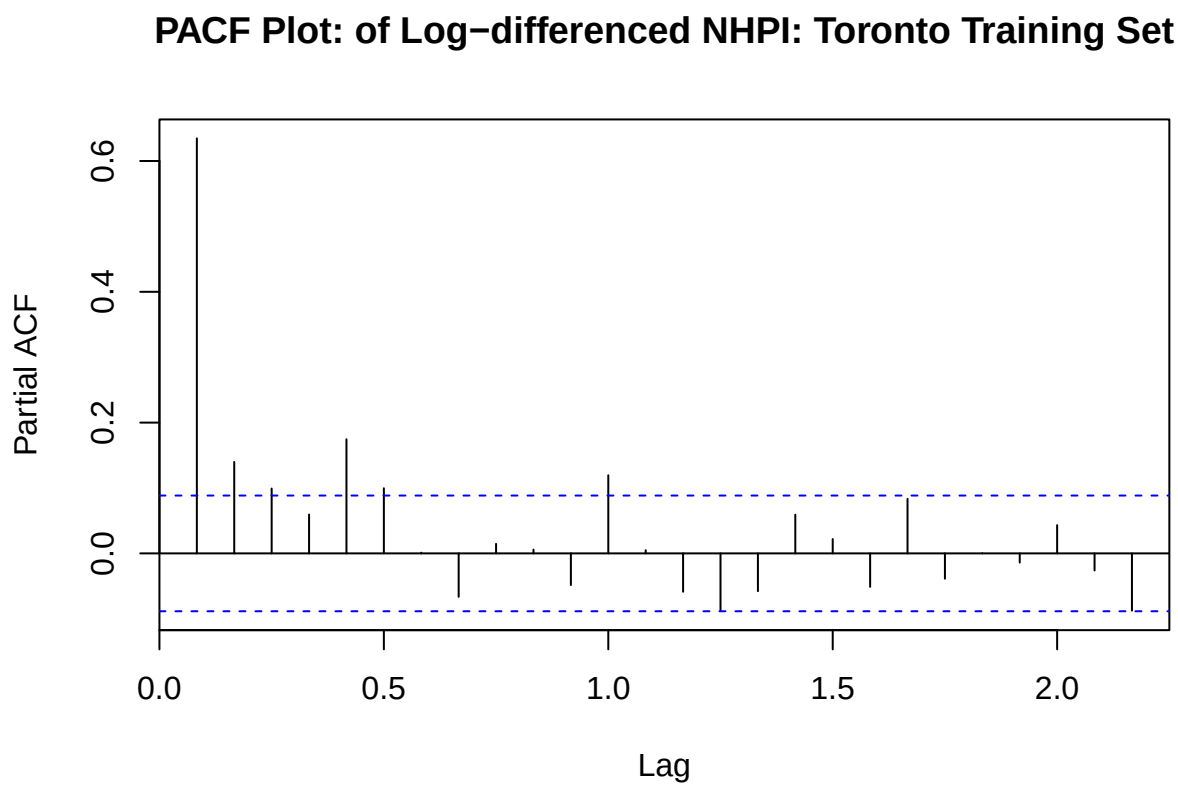

```r
# ts plot of holdout set
plot(ts(toronto_holdout_ts, start = c(2022, 1), frequency = 12))
```



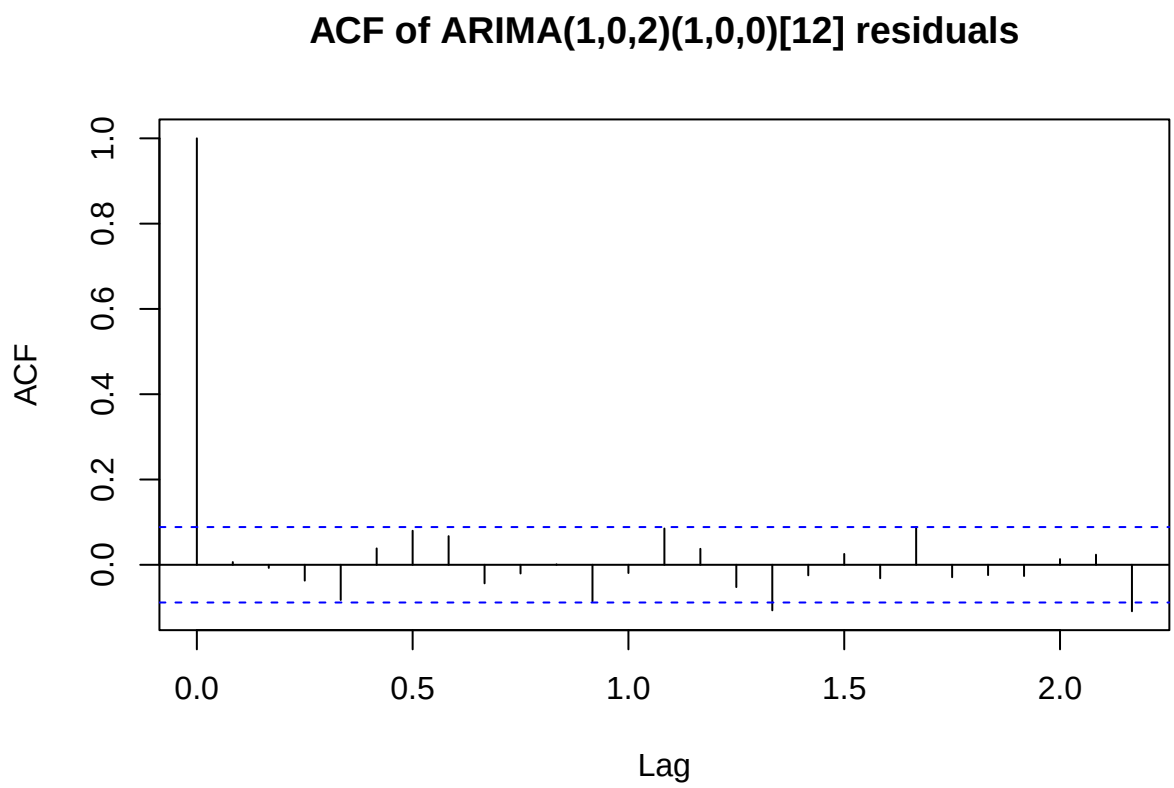

```
# 2. ACF PLOT
# ACF and PACF of training set
acf(toronto_train_ts, main = "ACF Plot: of Log-differenced NHPI: Toronto Training Set")
```



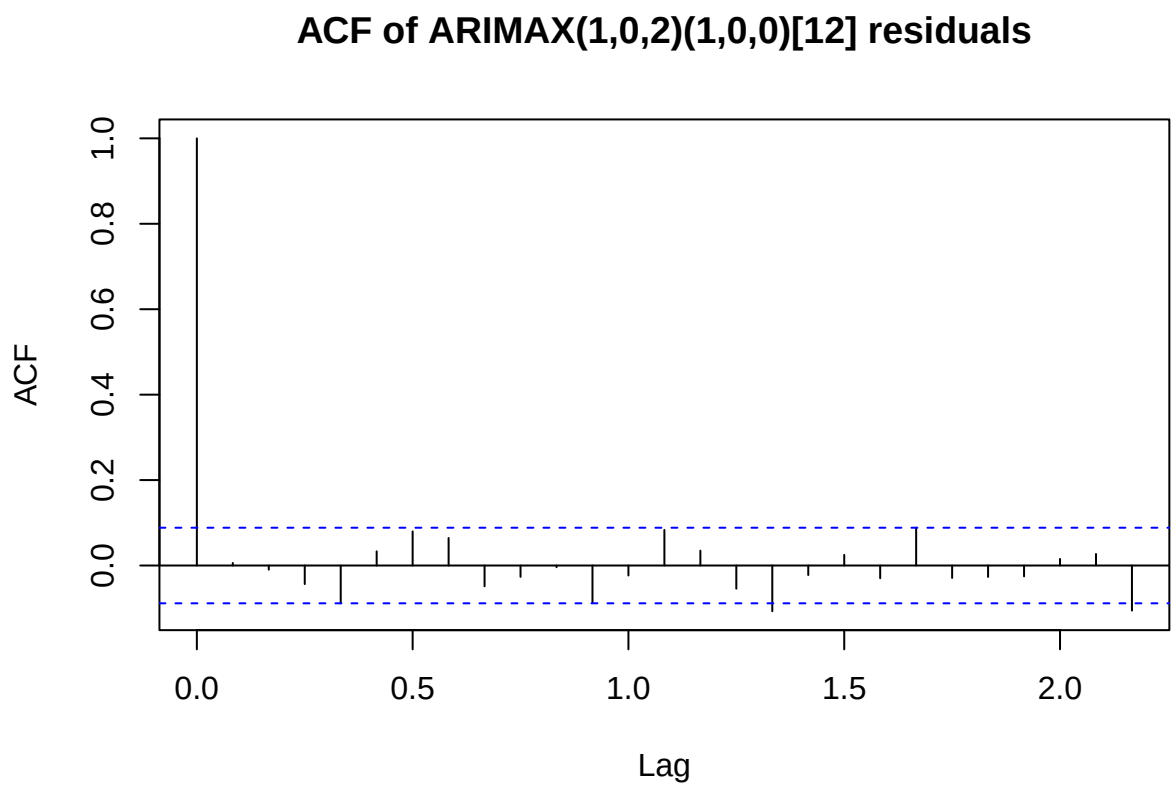

```
pacf(toronto_train_ts, main = "PACF Plot: of Log-differenced NHPI: Toronto Training Set")
```




```r
# ACF from best arima model
resid_fit <- residuals(toronto_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(1,0,2)(1,0,0)[12] residuals")
```




```r
# ACF from best arimax model
resid_fit <- residuals(toronto_fit_arimax)
acf(resid_fit, main = "ACF of ARIMAX(1,0,2)(1,0,0)[12] residuals")
```




```r
# 3. SUMMARY STATISTICS
summary(toronto_train_ts)
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -0.050552  0.000000  0.001820  0.002861  0.004292  0.061174
```

```r
summary(toronto_holdout_ts)
```

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.0034662 -0.0002171  0.0000000  0.0001822  0.0000000  0.0069445
```

```r
# summary of best ARIMA model
summary(toronto_transformed_nphi_auto)
```

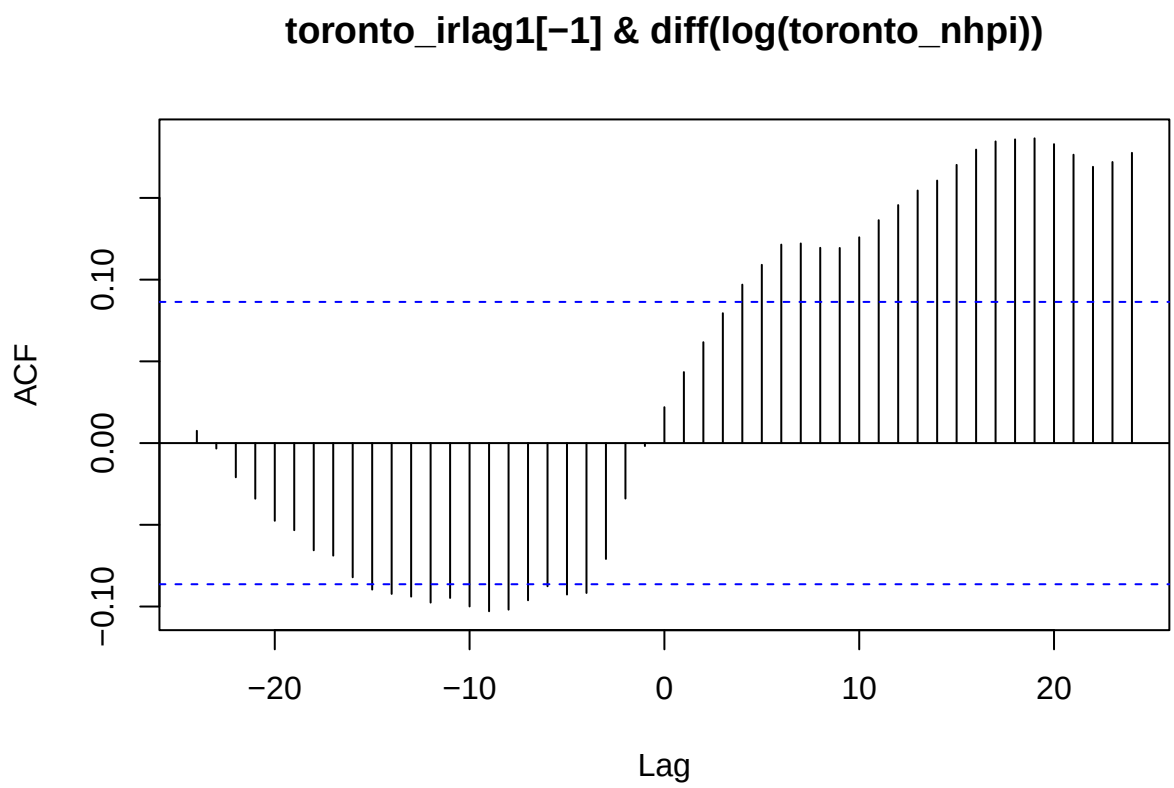
```
## Series: toronto_train_ts
## ARIMA(1,0,2)(1,0,0)[12] with non-zero mean
##
## Coefficients:
##          ar1      ma1      ma2    sar1    mean
##       0.9161  -0.4317  -0.1386  0.1316  0.0034
## s.e.  0.0302   0.0551   0.0490  0.0487  0.0017
##
## sigma^2 = 4.146e-05:  log likelihood = 1782.61
## AIC=-3553.23   AICc=-3553.05   BIC=-3528.05
```

```
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001299236 0.006405744 0.003868432 NaN  Inf 0.6292045
##                     ACF1
## Training set 0.006342322
```

```r
# summary of best ARIMAX model
summary(toronto_fit_arimax)
```

```
## Series: toronto_train_ts
## Regression with ARIMA(1,0,2)(1,0,0)[12] errors
##
## Coefficients:
##          ar1      ma1      ma2    sar1  intercept    xreg
##       0.9257  -0.4415  -0.1406  0.1330     0.0051  -3e-04
## s.e.  0.0329   0.0563   0.0488  0.0489     0.0028   4e-04
##
## sigma^2 = 4.147e-05:  log likelihood = 1783
## AIC=-3551.99   AICc=-3551.76   BIC=-3522.62
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001599343 0.006400238 0.003875959 NaN  Inf 0.6304288
##                     ACF1
## Training set 0.006084236
```

```r
# ccf between log-differenced NHPI and the lagged interest rate
ccf(toronto_irlag1[-1], diff(log(toronto_nhpi)))
```




```r
# 4. RMSE
rmse = cbind(toronto_persist_fc$rmse, toronto_iid_fc$rmse,
             toronto_esm_fc$rmse, arima_results$rmse, arimax_results$rmse)
colnames(rmse) = c("persist", "iid", "esm", "arima", "arimax")
print(round(rmse, 3))
```

```
##   persist   iid   esm arima arimax
## a   0.002 0.003 0.002 0.002  0.002
```